

**“Studies on variability and management of
Rhizoctonia bataticola (Taub.) Butler causing
dry root rot of chickpea”**

THESIS



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**Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya,
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By

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2018

CERTIFICATE - I

This is to certify that the thesis entitled “**Studies on variability and management of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea**” submitted in partial fulfillment of the Degree of **MASTER OF SCIENCE IN AGRICULTURE (Plant Pathology)** of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior is a record of the bonafide research work carried out by **Mr. JAGDISH PRASAD** under my guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma or has been published. All the assistance and help received during the course of these investigations has been acknowledged by the scholar.

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CERTIFICATE – II

This is to certify that the thesis entitled “**Studies on variability and management of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea**”. submitted by **Mr. JAGDISH PRASAD** to the **Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior**, in partial fulfillment of the requirements for the degree of **Master of Science in AGRICULTURE** in the **Department of Plant Pathology** has been accepted after evaluation by the external examiner and approved by the Student's Advisory Committee after oral examination.

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ABBREVIATIONS

S.N.	Legends	Description
1	@	At the rate of
2	%	Per cent
3	⁰ C	Degrees Celsius
4	ANOVA	Analysis of variance
5	C.V.	Coefficient of variation
6	cal.	Calculated
7	CD	Critical difference
8	Cfu	Colony forming unit
9	Cm	Centimeter
10	CRD	Completely Randomized Design
11	DAT	Day after transplanting
12	DF	Degrees of freedom
13	EC	Emulsifiable Concentrate
14	EMS	Error means sum of square
15	<i>et al.</i>	Allied (and other)
16	Fig.	Figure
17	G	Gram
18	gl ⁻¹	grams/ liter
19	i.e.	That is (in reference to)
20	Kg	Kilogram
21	mm.	Millimeter
22	MSS	Mean Sum of Square
23	No.	Number
24	PDA	Potato Dextrose Agar
25	Ppm	Parts Per Million
26	Q	Quintal
27	S.N.	Serial number
28	S.S.	Sum of Square
29	S.V.	Source of variance
30	SE (m)±	Standard error of mean
31	SL	Suspendable liquid
32	spp.& sp.	Species
33	tab.	Tabulated
34	t/ha	Tones/hectare
35	Viz.,	Wide list
36	WG	Wettable granule
37	WS	Water Soluble

CHAPTER-I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the world's third most important food legume after soybean and pea. Out of total production of pulses, chickpea accounts for 20 per cent. India contributes the major share to the global chickpea area (about 65%) and production (65%). In India, it is grown on 9.93 million ha area with a production of 9.53 million tonnes and productivity of 960 kg/ha (Anonymous, 2015). Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra, Karnataka, Bihar, Andhra Pradesh, Chhattisgarh, Haryana, West Bengal are the Chickpea growing areas of India. Among the chickpea growing states, Madhya Pradesh is one of major chickpea growing state in India, which covers the largest area of 3.16 million ha and contributes about 3.29 million tones with an average productivity of 1044 kg /ha (Anonymous, 2015).

Chickpea is the best and cheapest source of protein in human diet. It is the rich source of protein (21.1%) carbohydrate (61.5%) and fat (4.5%). Chickpea is mainly used for human consumption as well as animal feed. It is consumed as whole seed, fried, boiled, salted or more generally which is cooked. Fresh green leaves are used as vegetable. Exudation from leaves (locally called *amb*) contain malic acid and oxalic acid which possess medicinal value for bronchitis, diarrhea, digestive disorder, warts and blood purification. Germinated seed are recommended to cure scurvy disease.

Chickpea productivity often remains low due to susceptibility of the crop to different biotic and abiotic stresses. Among the biotic stresses, diseases, insect pest, nematodes and parasitic weeds account for major losses. The crop is attacked by 172 pathogens which include 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.*, 1984).

Among the disease dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler (pycnidial stage: *Macrophomina phaseolina* Tassi Goid) is emerging as serious threat to the chickpea growing regions of India (Masood Ali and Shivkumar, 2001). The pathogen survives in soil on infected host

debris. Secondary spread of the disease is through farm implements, irrigation water and rain splash. Recently wilt resistant chickpea cultivars were found succumbing to dry root rot in major growing areas (Pande *et al.*, 2010). Being soil borne in nature, it is very difficult to control this pathogen. Yield losses due to root rot are more when the conditions are congenial for disease development. Losses to the extent of 70.8 per cent at full podding and 48.9 per cent at preharvest stage of the crop have been reported due to the disease (Ahmed and Mohamad, 1986).

R. bataticola is a diverse, omnipresent soil borne fungal pathogen with a wide host range involving more than 500 species in 75 families (Dhingra and Sinclair, 1978). It mainly produces macro-sclerotia (primary source of inoculum) or pycnidia. It is both soil and seed borne and infects plants mainly from flowering stage to maturity. The disease is reported to be more severe when chickpea crop is exposed to moisture stress conditions (Sharma *et al.*, 2010). Natural population of this pathogen is diverse in terms of pathogenic variability and can be characterized into pathotypes or races by means of biological typing. Previous studies have revealed and reported high levels of pathogenic and genetic variation in *R. bataticola* from different parts of the world and India (Trivedi and Gurha, 2006; Aghakhani and Dubey, 2009; Pande *et al.*, 2010).

Considering the importance of the disease and area under chickpea cultivation in Madhya Pradesh, the present investigation was undertaken to work out the variability in *R. bataticola* isolates from chickpea growing areas of Sehore district of Madhya Pradesh with following prime objectives-

1. Survey of dry root rot disease of chickpea in and around Sehore.
2. Isolation of causal organism of dry root rot of chickpea and its pathogenicity tests.
3. Morphological and pathogenic variation among isolates of *R. bataticola*.
4. Effect of bio agents and fungicides on growth of chickpea plant and their disease management efficiency.

CHAPTER II

REVIEW OF LITERATURE

A brief review of literature pertaining to occurrence and distribution of dry root rot disease, symptomatology, causal organism responsible for the disease and its cultural, morphological, pathological variability among the isolates, effect of environmental factors on disease and management of by fungicide and bio agents were presented in this chapter.

Occurrence and distribution of disease

Dry root rot chickpea was first reported from India by Mitra (1931), later, the disease has been reported from most chickpea-growing areas in India like Punjab by Luthra (1938), Madhya Pradesh (Kotasthane *et al.*, 1979), Uttar Pradesh (Pandey and Singh, 1990). It has been reported from other countries like Iran (Kaiser *et al.*, 1968), the USA (Westerlund *et al.*, 1974) and several countries in Asia and Africa (Nene *et al.*, 1996). The disease was earlier known as “*Rhizoctonia* wilt” in chickpea; however, later it was named as “dry root rot”.

The incidence of dry root rot in chickpea caused by *R. bataticola* was studied in 20 locations in Uttar Pradesh and Uttaranchal, India during February to March of 1998-2000. The disease was widespread in Uttar Pradesh and it was severe in bold seeded cultivars (Prajapati *et al.*, 2003). Shubha and Gurha (2006) found 5-22% infection of *R. bataticola* on chickpea in Bhundelkhand region of Uttar Pradesh.

Gurha and Trivedi (2008) reported 60-70% incidence of dry root rot in chickpea fields of Shimoga, Raichur and Bangalore districts in Karnataka.

Manjunatha *et al.* (2011) reported maximum dry root rot incidence in chickpea in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18 %) in Bidar (Karnataka). The overall incidence of dry root rot ranged from 1-19% across the districts. This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rain fed condition.

Survey of Jammu and Kashmir showed highest incidence up to 40% in Shangus and lowest up to 4.11% in Naina. The incidence of dry root rot in chickpea caused by *R. bataticola* was observed in late October to mid

November. The intensity of the disease was high in the month of February and March during late flowering and podding stage (Khan *et al.*, 2012).

Recent surveys conducted during 2010-2013 indicated widespread and increased incidence of dry root rot in the central and southern states of India (Ghosh *et al.*, 2013). Disease was found irrespective of soil types, cropping system and cultivars used and incidence ranged from 5 to 50% or more in badly infected soils. This noticeable widespread geographic distribution of dry root rot probably makes it a significant disease in future.

The dry root rot of was not of much significance in chickpea earlier, but now, it has become a major threat to chickpea production in recent years due to altered weather conditions, particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at post-harvesting stage is predisposing chickpea to dry root rot (Sharma and Pande, 2013).

Disease symptomatology

The dry root rot symptoms were observed in chickpea during post flowering stage showing bronzing of the leaves on one or more of the lower branches (Dastur, 1935). The colour of leaves later changed to yellow and then brown. The affected branches and leaf stalks of diseased plants were stiff and turned upwards and leaflets stand more or less vertically and shed prematurely.

The tap root turns black with sign of rotting and is devoid of most of the lateral and tiner roots. The dead root are quit brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil, when plants are uprooted. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark or when split open at the collar region vertically (Sharma *et al.*, 2015).

Singh *et al.* (1990) reported that roots inoculated with *R. bataticola* (*M. phaseolina*) caused disintegration of cortical tissues with mycelial and scleritial bodies plugging the xylem vessels.

Pathogen

Rhizoctonia is a genus of anamorphic fungi in the order Cantharellales and family Ceratobasidiaceae. *Rhizoctonia* species do not produce spores, but are composed of hyphae and sclerotia (hyphal propagules), asexual stage

of fungi. *Rhizoctonia* species are saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de Candolle for plant pathogenic fungi that produce both hyphae and sclerotia. Subsequent authors added over 100 additional names to the genus. A comprehensive survey and redistribution of species name “*Rhizoctonia*” was published in 1994 by Anderson and Stalpus. According to this published information, *Rhizoctonia bataticola* is used as a synonym to *Macrophomina phaseolina* (tassi) Goid 1947. Currently, *M. phaseolina* is officially recognized as the correct taxonomic name. The sclerotial phase known as *R. bataticola* (Holliday and Punithalingam 1970).

Mycelia width varied from approximately 2-11µm and distance between two consecutive septa measured 46µm. However, the most important character regarding taxonomy and classification were the production size and composition of microsclerotia (Reichert and Hellinger, 1947).

Sclerotia within plant parts were black, smooth, hard and varied in size from 100 µm-1mm while in culture, it varied from 50-300µm. These descriptions were given by Commonwealth Mycological Institute (CMI), Kew, England (1970). During the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies called microsclerotia. The microsclerotia were black and variable in size from 50–150 µm depending on the available nutrients of the substrate on which the propagules were produced (Short and Wyllie, 1978).

Under *in vitro* conditions, the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates, or mycelium is found to be completely or partially suppressed (Sharma *et al.*, 2012). The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes (Sharma *et al.* 2012)

Variability in *Rhizoctonia bataticola*

Much work has been done to elucidate the variability in morphology, physiology, pathogenicity and genotypes of *M. phaseolina* in different hosts [Dhingra and Sinclair, 1973 (castor); Anil kumar and Sastry, 1982 (sunflower); Sobti and Sharma, 1992 (groundnut); Ratnoo *et al.*, 1997 (cowpea); Atiq *et al.*, 2001 (sunflower); Okwulehie, 2001 (groundnut); Suriachandraselvan and Seetharaman, 2003 (sunflower); Sharma *et al.* 2004 (pearl millet, sesame, horsegram and mothbean); Fernandez *et al.*, (2006) (bean); Ndiaye, 2007 (cowpea)]. The pathogenic variability in *R. bataticola* is assumed to be due to mutation, hyphal fusion and mitotic recombination.

Hyphal variability

Sobti and Sharma (1992) reported that the width of hyphae varied from 4.16-8.48µm in *R. bataticola* isolates from groundnut.

Devi and Singh (1998) studied the morphological variability of thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India. They found typical right-angled branching of mycelium in isolate MP-1 and acute to right angled in isolates MP-2 and MP-3.

Pathogen was isolated from stem and leaves of black gram plants by Suryawanshi *et al.* (2008). The mycelium was superficial, hyaline to brown, septate and tree like in form.

However, not much has been reported on *R. bataticola* of chickpea. Sharma *et al.*, (2012) reported cultural, morphological and molecular variations in 94 isolates of *R. bataticola* collected from various chickpea grown in various agro-ecological zones of India. The isolates varied in cultural and morphological characters like colony colour, growth pattern, growth rate and sclerotial initiation and production.

Gupta *et al.* (2012) observed variation in size of hyphal width of forty *R. bataticola* isolates from chickpea. Eighteen highly virulent isolates were classified under very thin (5.2-6.5 µm), 17 highly virulent and 2 virulent isolates under thin (6.6 -7.9 µm) and only three highly virulent isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 µm) thick hyphal width.

Sclerotial variability

Hildebrand *et al.* (1945) observed black, smooth sclerotia varying from spherical to oblong to irregular shaped. The sclerotial shape of isolate varied with substrate on which fungus was grown. Sclerotial diameter of *R. bataticola* in gram varied from 98.1-119.2 μm (Ghosh and Sen, 1973).

Morphological variability among thirty six isolates of *M. phaseolina* from mung bean collected from different region of North, South, North East and Central India were studied for by Devi and Singh (1998). They observed bigger sclerotia in isolate MP-2 (400 x 280 μm).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their morphological characters. The sclerotia formed in different isolates were dark brown to black and size varied from 40 - 600 μm . The isolates from Bangalore (Rb1) and Faridkot (Rb 5) produced the largest size sclerotia ranged from 200 – 600 μm and 100 – 400 μm respectively. Dumka (Rb19) isolate produced small sized sclerotia which ranged from 43.9- 81 μm . Most virulent isolate Rb1 from Bangalore, (Karnataka) was fast growing and produced largest sclerotia. Isolates were highly variable in virulence (Aghakhani and Dubey, 2009).

Manjunatha and Naik (2011) collected thirty isolates of *R. bataticola* causing dry root rot of chickpea from three major chickpea growing regions variable in their characters were categorized into three groups. The first group comprised of all isolates from Bidar except RB7 and exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur and Gulbarga except RG3 and RG7 and also an isolate from Bidar RB7, showed brownish black flat sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown coloured sclerotia formed at the center of the culture.

In chickpea, Sharma *et al.* (2012) found significant relationships between sclerotial initiations, sclerotial, intensity and disease severity between isolates. Positive correlation between pathogenic isolates and

sclerotial production was found in mung bean (Hooda and Grover 1988). However, Manici *et al.* (1992) found no such correlation in sunflower. Than *et al.* (1991) studied the relationship among *R. bataticola* isolates based on colony fusion type with 37 isolates in chickpea and found existence of more than one isolate at different places. On the basis of virulence study of 40 isolates of *R. bataticola* on chickpea- susceptible cultivar BG 212, two pathotypes were reported from central region in India (Gupta *et al.*, 2012).

Eleven isolates of *R. bataticola* from different pulse crops viz., redgram, greengram, cowpea, soybean, blackgram showed variability in sclerotial intensity, shape and intensity of pigment synthesis. All isolates were more aggressive on the original host from which it was isolated than on other hosts (Sundravadana *et al.*, 2012).

Isolates of *M. phaseolina* were collected from semi arid regions of Maharashtra, Andhra Pradesh and Karnataka states of India. Variability in shape, length, width and length/width of microsclerotia were studied. AP 21 isolate varied in length/width ratio (483.3/200.0 μm) while no significant differences were observed in others (Prasad *et al.*, 2014).

Cultural variability among *Rhizoctonia batatiocla*

Dhingra and Sinclair (1973) collected nine isolates of *M. phaseolina* from various parts of United States and reported variation in growth rate and colony characters. Further, the growth rate was correlated with pathogenicity.

Lokesha and Benagi (2004) reported variation among four isolates of *M. phaseolina* in pigeonpea. Maximum growth was recorded in ICRISAT isolate which was significantly superior over other isolates (82.7 mm) followed by Bidar isolate (78.35 mm).

The radial growth was found maximum after 24 hours of incubation on PDA (40.3 mm) followed by potato sucrose agar (39.0 mm) and it completely covered the PDA plate by 48 hours of incubation (Chowdary and Govindaiah, 2007).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing of India were highly variable in their cultural characters. After 48 hours of incubation, highest radial growth in Rb1 (Bangalore) followed Rb4 (Faridkot) was observed. Rb3 (New Delhi) and Rb5 (Faridkot) were next best with statistically similar colony

diameters (83.5 mm). Least and similar colony diameter of 51.3 mm was recorded in Rb7 and Rb10 (Kanpur) isolates (Aghakhani and Dubey, 2009).

Iqbal and Mukhtar (2014) collected 65 isolates of charcoal rot caused by *M. phaseolina* in mung bean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan, were studied for cultural variability. Significant differences were noticed among 65 isolates, sixteen isolate rated as fast growing, 11 as slow growing and the rest as medium growing. Maximum colony diameter of 87.17 and 86.67 mm was observed in isolate MP- 7 (Dera Ghazi Khan) and MP-26 (Layyah) proved to be the fast growing, while isolates MP-8, MP-29 and MP-30 showed minimum radial growth and rated as slow growing.

Variability in colony texture

Dhingra and Sinclair (1978) noticed fluffy growth in stem isolates, partially fluffy growth in root isolates and appressed growth in all other plant part isolates of *M. phaseolina* on soybean. MP-1 isolate of mungbean had cottony mycelial growth while MP-2 isolate showed 5 μ m appressed mycelial growth (Devi and Singh, 1998).

The diversity patterns of 15 isolates of *M. phaseolina* from Mexico and 15 isolates from other countries on beans were analyzed on the basis of presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth (Fernandez *et al.*, 2006).

Aghakhani and Dubey (2009) reported high variability in colony texture among thirty three isolates of *R. bataticola* (*M. phaseolina*) causing dry root rot of chickpea. The majority of nineteen isolates produced suppressed mycelium whereas four isolates in which three from Kanpur and one from Rohtak produced aerial mycelium.

Sundravadana *et al.* (2012) reported variability among the eleven isolates of *R. bataticola* from different pulse crops (redgram, greengram, cowpea, soybean, blackgram). They categorized it into three groups viz., linear, fluffy and linear at the end with fluffy growth at the centre. All isolates were more aggressive on the original host from which it was isolated.

pigmentation

Devi and Singh (1998) studied the colony colour of thirty six isolates of *M. phaseolina* of mung bean. MP-1 isolate had pale grey colonies while brown in MP-2.

Out of 30 isolates of *M. phaseolina* of beans, 22 isolates were grey, 4 isolates were white and 4 isolates were black in colour (Fernandez *et al.*, 2006).

Aboshosha *et al.* (2007) observed colonies of *Macrophomina* isolates of sunflower charcoal rot were mostly grey, dark grey, black and brown on PDA and Czapek-Dox Agar. White colony was observed in only one isolate on both the media.

Pigmentation of twenty three isolates of *R. bataticola* varied from white to dull white, creamy, grey and black. Black was prominent among the isolates (Aghakhani and Dubey (2009).

Pathogenic variability

Pre-germinated seed of 21 chickpea varieties were sown in soil inoculated with *R. bataticola*. Seed rot were observed within 24h in NEC874 and EG234 while Bold 2375, BG209, JG62, JG315, ICC 3357 and JG1133 developed necrotic lesion in 3-5 cells deep at hypocotyl region within 7 days. RSG-44, AGC677, NEC41, GL269, JG74, ICC8983 and ICC 5003 developed only superficial necrosis along the hypocotyl region. BGM 416, BG 416, ICC 1376 and ICC 113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases Jayanti and Bhatt (1993).

Pande *et al.* (2004) reported resistance to dry root rot in germplasm lines ICC 14395, cultivar ICCV 2 and advanced breeding line (ICCX830203-BH-BH-11H).

Jayalakshmi *et al.* (2008) screened 12 promising entries of chickpea (JG-11, GBS 964, GBS 963, GBC 2, GBM 6, GBM 2, GBM 10, BGD 103, GCP 107, GCP 101, A1, ICCV 10, L- 550 and 1 susceptible check JG-62) against dry root rot during rabi season, 2004-05 and 2005-06. Out of 12 genotypes, four were resistant, two moderately susceptible, three susceptible and three were highly susceptible to dry root rot disease.

Lekhraj *et al.* (2012) found pathogenic variability on chickpea infected with *M. phaseolina* isolates collected from 9 various parts of Rajasthan, Hissar

and New Delhi. Pathogenicity of these isolates on 40 day old chickpea cv. C-235 showed no direct relationship between cultural characters and virulence of the isolates. The isolate BKN had scanty, suppressed growth and was highly pathogenic. Isolates SGN, NHR, DUR, HAU, DEL and HNG having different types of colony growth were moderately pathogenic and the isolates CHR and UPR with different mycelium characters were less pathogenic. Similarly, there was no apparent correlation between the virulence and growth rate of the fungal colonies.

Fifty isolates of *R. bataticola* collected on chickpea from different agro climatic zones in India proved pathogenic on chickpea cultivar BG 212. The dry root rot disease severity ranged from 4 to 9 (1-9 severity rating scale). The effect of isolates on severity was significant. The pathogenic and non pathogenic isolates were not concentrated in any one particular state/region (Sharma *et al.* 2012).

Influence of environmental factors on disease development

Dry root rot was observed in gram when exposed to temperature more than 30°C. The disease was aggravated by dry soil conditions especially at flowering stage leading sudden drying of plants (Singh and Mehrotra, 1982).

Raabe (1985) found that *R. bataticola* causes severe damage to chickpea in the warmer Salinas Valley in California as the pathogen is becoming more virulent under high temperatures (32°C). *R. bataticola* can grow and produce large amounts of microsclerotia under relatively low water potentials allowing the disease on bean to be recognised as favouring drought (Olaya and Abawi, 1996). The survival of sclerotia of *R. bataticola* in black soil was affected by high soil moisture (80%), while the saprophytic activity of pathogen was maximum at low moisture levels (20,30 and 40%) (Maheswari and Ramakrishnan, 1999). The adverse effect of high soil moisture on survival of sclerotia of *R. bataticola* was also reported by Olaya and Abawi, (1996) indicating that *R. bataticola* cannot survive for a longer period under anaerobic conditions. Similar to chickpea, in soybean and cowpea, *M. phaseolina* population in the soil was negatively correlated with soil moisture and positively with maximum soil temperature. (Gupta and Gupta, 1986). The dry root rot incidence of chickpea was maximum in sandy soils (86.6%) as compared with clay soils (61.6%) as observed by Taya *et al.* (1988) in sick

pots at greenhouse. The deficit soil moisture was generally detrimental to both chickpea plant growth and pathogen activity (Bhatti and Kraft, 1992).

Colonization of *M. phaseolina* on chickpea roots was high when subjected to moisture stress conditions compared to unstressed plants. Soils artificially inoculated produced poor growth of plants both in low and high moisture compared to natural soil. Root colonization was greater at 40-50% moisture holding capacity as compared to 10-20% soil moisture holding capacity (Husain and Ghaffar, 1995).

Patel and Anahosur (2001) conducted an experiment with chickpea cultivars ICC 4951 (susceptible) and Bheema (resistant) in earthen pots filled up with artificially infected soil with *M. phaseolina* at four levels of soil moisture viz., 25, 50, 75 and 100%. They observed *M. phaseolina* infection was maximum at 25% and minimum at 100% soil moisture. ICC 4951 developed higher disease than Bheema at 25% soil moisture. Both cultivars showed maximum disease incidence at the lowest soil moisture level compared to high moisture.

High day temperature above 30°C and dry soil conditions at flowering and podding stages rapidly increase the severity of the dry root rot disease on chickpea (Gurha *et al.*, 2003). Singh and Sharma (2002) observed that deficit soil moisture favours the severe disease development on pulse crops. Under hot and dry environmental conditions many economically important crops are predisposed to the infection and colonisation of *R. bataticola* and caused drastic yield losses on chickpea (Thripathi and Sharma, 1983), soybean (Pearson *et al.*, 1984, Wokocha, 2000) and sunflower (Nawaz, 2007).

With increasing temperature and more frequent moisture stress, dry root rot is becoming more intense in typically tropical humid areas. Increasing incidence of dry root rot at various locations over years suggests a strong influence of rising temperatures on *R. bataticola* (Savary *et al.*, 2011). Data collected in India from 2000 to 2010 showed higher incidence of dry root rot in chickpea varieties that are resistant to Fusarium wilt in years when temperatures exceed 33°C at flowering to podding stages (Pande *et al.*, 2010; Sharma *et al.*, 2010). Under controlled environment Sharma and Pande (2013) observed that *R. bataticola* infected chickpea plants had caused dry root rot faster at 35°C with soil moisture levels less than or equal to 60%.

Further, *R. bataticola* propagules were negatively correlated with high soil moisture and positively with less soil moisture content.

Management of disease

Biological control

Seed treatment with biocontrol agents like *Trichoderma viride* has shown some benefits in managing the dry root rot of chickpea (Singh *et al.*, 1998; Indira and Thiribuvanamala, 2002; Gurha *et al.*, 2003). Antagonistic effect of four bacterial and six actinomycetes isolates on *R. bataticola* of chickpea was reported by Singh and Mehrotra (1982). Biological control of *R. bataticola* of black gram was most effectively controlled by the integration of antagonist (*Trichoderma virens*) and organic amendments (FYM) and significantly improved the plant growth, grain yield, dry matter production and reduced disease incidence than application of *T. virens* alone (Christopher *et al.*, 2007). A combination of biocontrol agents of *T. viride*, *Pseudomonas fluorescens* and *Bacillus subtilis* improved the management of *R. bataticola* in mung bean (Thilagavathi *et al.*, 2007).

Manjunatha and Naik (2010) evaluated isolates of two bio-control agents *T. viride* and *P. fluorescens* against *M. phaseolina* under laboratory conditions and field. The most effective isolate of each bio-control agent and the common chemical for seed treatments were too used as experimental trail. It was found that trial combining with soil application through bioagent enriched farm-yard manure, along with seed treated with the bio-control agent showed maximum germination, least root rot incidence and highest yield, as compared to the other biological or chemical seed treatments included.

Govindappa *et al.* (2010) isolated three biocontrol agents *T. harzianum*, *P. fluorescens* and *Bacillus subtilis* from safflower rhizosphere soil and tested individually for their effectiveness in controlling root-rot of safflower. Talc based formulations of *P. fluorescens* and *T. harzianum* (10 g/kg) proved to be effective in controlling disease under laboratory, greenhouse and field conditions. The efficacy of these biocontrol agents are equivalent to the standard fungicide Bavistin (carbendazim). Seed treatment with these biocontrol agents enhanced the seed germination, growth parameters and root-rot disease.

Jyothirmai *et al.* (2011) found talc based formulations of *Trichoderma* spp. and fluorescens *Pseudomonas* isolate 1, effective in suppressing the seedling disease and improved the plant growth, shoot and root dry weight and pod yield.

In the field trial seed treatment with four strains significantly reduced the disease incidence of dry root rot in all treatments in comparison to uninoculated control, however *P. fluorescence* strains BHUPf4 was found more effective in mycelial growth reduction as well as in disease reduction followed by strains BHUP5, BHUP6 and BHUPsb (Patel *et al.*, 2011).

Govindappa *et al.* (2011) screened thirty eight isolates of *P. fluorescens* for biological control of *M. phaseolina* root-rot of safflower. Results indicated that the seedling assay is more important for selection of promising biocontrol agents. *P. fluorescence* also increased the plant growth besides controlling the disease.

Kumar *et al.* (2013) evaluated three bio agents, *T. harzianum*, *T. Viride* and *T. atroviride*, and one bacterium, *Bacillus subtilis* isolated from moth bean against eight isolates of *M. phaseolina* causing dry root rot of moth bean. Inhibition of the mycelium growth of *M. phaseolina* isolates by *T. harzianum*, *T. viride*, *T. atroviride* and *B. subtilis* varied from 61.1 to 70.1%, 58.6 to 66.6%, 52.0 to 63.1% and 45.8 to 54.8%, respectively. Maximum growth inhibition was caused by *T. harzianum*. In a greenhouse study, these antagonists reduced root rot incidence in a susceptible moth bean cv. Rmo-225 up to 69.0, 65.0, 62.5 and 58.3%, respectively, as compared to control where 100 % mortality was observed. *T. viride* was most effective among the antagonists trial in minimizing the disease followed by *T. harzianum*, *T. atroviride* and *B. subtilis*.

Nagamani *et al.* (2015) reported that seed treatment with carbendazim and *T.viride* along with soil application of FYM (Farm Yard Manure) fortified with *T. viride* as the most effective in decreasing the mortality per cent of dry root rot of chickpea.

Marisa *et al.* (2014) evaluated *Pseudomonas chlororaphis* sub spp. *aurantiaca* for control of *M. phaseolina*. The bio agent caused a significant inhibition of *M. phaseolina in vitro* and reduced damping-off in the *in vivo* assays.

Chemical control

Seed treatment by fungicides is effective to some extent in reducing losses caused by *R. bataticola* in crops which are particularly vulnerable at the seedling stage. Fungicide seed treatment with carbendazim and thiophanate methyl and vitavax reduced the dry root rot of chickpea significantly over untreated check (Taya *et al.*, 1990; Bhardwaj, 1995; Singh and Sindhan, 1998; Rathore and Rathore, 1999; Sharma and Gupta, 2004). The combined use of host resistance with fungicide treatment resulted in better seedling emergence and delayed the onset of root rots.

Pardeshi *et al.* (1989) reported that root and shoot lengths and seedling fresh weight were increased most effectively after application of captan, followed by thiram, sulphur and mancozeb. Most of the fungicide treatments increased seedling dry weight.

Treating the seeds with captan or thiram is also helpful in reducing the disease (Gurha *et al.*, 2003). Practice of seed treatment with fungicide (bavistin and thiram) in farmers' field reduced the dry root rot incidence in central and southern parts of India (Ghosh *et al.*, 2013). Taya *et al.* (1990) in greenhouse tests found the best control of *R. bataticola* was given by carbendazim alone or in combination with thiram as seed treatment, pre-sowing soil-drench and seed treatment + drenching after sowing.

Vijay-Mohan *et al.* (2006) reported management of dry root rot by using carbendazim (0.2%) and etaconazole (0.1%) as seed treatment, soil drenching and seed treatment plus soil drenching.

Thawale *et al.* (2010) found that seed treatments with thiram + carboxin and hydration (2 hrs) followed by dry dressing with thiram, recorded maximum germination, seedling length, seedling dry weight and vigour index. All treatments recorded more number of pods/plant, 100 seed weight, and seed yield/plant over control.

Manjunath, *et al.* (2011) evaluated two biocontrol agents individually and in combinations and in integration with carbendazim seed treatment and hexaconazole seed soaking for management of dry root rot of chickpea. Seed and soil application with bioagents was effective for the maximum germination with less root rot incidence as compared to seed treatment with carbendazim alone and seed soaking in hexaconazole alone. Integration of seed and soil

application of biocontrol agents resulted in higher germination and reduced mortality due to disease as against check. The application of biocontrol agents individually either soil application or seed treatment also showed good germination with reduced mortality due to dry root rot as compared to treatments imposed with fungicides.

CHAPTER-III

MATERIAL AND METHODS

(A) Materials

1. General

Corning and Borosil make glasswares were used throughout the course of study. Glasswares were cleaned with chromic acid solution followed by washing with detergent and finally rinsed with tap water. After drying, glasswares were sterilized in hot air oven at 180°C for 2 hours. The metallic equipments like forceps, needle and cork borer were sterilized by dipping in alcohol and heating to red hot over flame of a spirit lamp.

Surface sterilization of plant parts and diseased materials were done by dipping then in 0.1% mercuric chloride solution for 1-2 minutes and washed in sterilized water for 3 times.

The culture media were sterilized in autoclave at 15 lbs pressure per square inch (1.05 kg/cm²) for 20 minutes. The soil and sand were sterilized at 30 lbs pressure per square inch (3.1 kg/cm²) for two hours.

1.1 Media

Following media were used in the course of investigation with following ingredients:

1.1.1 Potato Dextrose Agar (Riker and Riker, 1939)

Peeled and Sliced Potato	200 g
Dextrose	20 g
Agar	20 g
Distilled water	1000 ml

1.1.2 Soil maize media

Maize powder	200g
Sand	400g
Distilled water	1000 ml

1.2 Source of seed and other materials

Chickpea seeds were procured from All India Coordinated Research Project on Chickpea, R.A.K. College of Agriculture, Sehore (M.P.). Fungicides, talk based formulation of *Bacillus subtilis* var. *amyloliquefaciens*

strain FZ824 (Taegro) and *Trichoderma viride* were obtained from the Department of Plant Pathology, R.A.K. College of Agriculture, Sehore (M.P.).

1.3 Survey

Survey for dry root rot was done in the chickpea growing areas of Sehore district. Diseased plant parts were collected in paper envelopes and brought to the laboratory for isolation. The chickpea crop was surveyed at flowering stage. During the survey per cent disease incidence was recorded.

1.4 Experimental site:

All the field experiments were conducted at the experimental field of Plant Pathology Farm of R.A.K. College of Agriculture, Sehore during rabi season 2013-14.

(B). METHODS

2.1 Collection of diseased plant samples and symptomatology

Dry root rot chickpea plants were collected from 28 location in Sehore district of Madhya Pradesh at adult plant stages. Samples were brought in the laboratory and symptoms were recorded in detail. The tissue of affected plants showing the symptom of dry root rot were examined under the microscope for the presence of the causal agent.

2.2 Isolation of pathogen from diseased samples

The roots of diseased plant showing symptoms were washed thoroughly with tap water, small pieces of infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl₂) solution for one minute followed by three washings with sterilized distilled water to remove traces of HgCl₂. The pieces were then transferred aseptically to Petri plates containing PDA. Inoculated Petri plates were incubated at 25± 2⁰C for three to five days and examined at frequent intervals to see the growth of the fungus developing from different pieces. As and when fungal colony appears they were transferred to PDA slant.

2.3 Purification and identification of isolates of *R. bataticola*

The cultures of *R. bataticola* were purified by frequent sub culturing by the hyphal tip method as described by Dhingra and Sinclair (1985). Isolated

fungus was identified their morphological characters according to Barnett and Hunter (1986) and stored at 4 °C until use.

2.4 Variability studies

2.4.1 Morphological and Cultural variability

Isolates of *R. bataticola* were grown on PDA (Potato Dextrose Agar) media in order to study their growth characters. Twenty ml PDA was poured in each of the previously sterilized petriplates and inoculated with five mm diameter disc of each fungal isolates at centre. Three replications were maintained for each isolate. The plates were incubated at 28 ± 1 °C for 7 days. The diameter of the mycelial growth was measured after 7 days. The type of mycelial growth was noted as Suppressed and Arial along with the pigmentation of culture on both upper and lower surface of petri plate observations of size and colours of sclerotia each of isolate were recorded.

2.4.2 Pathogenic variability

Pathogenic behavior of *R. bataticola* isolates were studied by soil infestation method. The *R. bataticola* isolates were multiplied on soil maize medium for 7 days. Sterilized soil was used in pot experiment. The soil was collected from the local field and dried thoroughly in sun and sterilized. Pots of 25 cm face diameter were used during experiments.

Trays were filled with sterilized soil and then 15 days old inoculum of each isolate of *R. bataticola* was added @20g/kg soil. The contents were then mixed and trays were left for 7 days to allow the multiplication as well as invasion of the fungus in the soil. In control, no inoculum was added and trays were filled with sterilized soil.

Seeds of ten varietie of chickpea viz., RVG-202, JG-16, RVG-203, BGD-112, JKG-1, L-550, Annegiri, JSC-37, JG-62 and JG-11 were sown in each isolates tray. Two trays were used for each isolates and all experiment was repeated twice.

The observation on germination was recorded after 10 days of sowing, while seedling mortality was recorded at fortnightly interval starting at 20 days after sowing (DAS).

The dead plants were observed critically for the association of *Rhizoctonia bataticola*. Isolations were made from the affected plants and the culture was compared with original one.

$$\text{Mortality percentage} = \frac{\text{No. of dead plants}}{\text{Total no. of observed plants}} \times 100$$

2.5 Effect of environmental factors on dry root rot of chickpea

Influence of rainfall, maximum and minimum temperature and relative humidity was examined on the development of dry root rot in the field.

The disease intensity was recorded in ten plants tagged randomly, at an interval of seven days till maturity of the crop.

The mortality percentage was calculated by applying the following formula.

$$\text{Mortality percentage} = \frac{\text{No. of dead plants}}{\text{Total no. of observed plants}} \times 100$$

The absolute disease intensity was calculated as the difference between two successive observations of mortality percentage and correlation coefficient ('r') was computed between the absolute disease intensity and different meteorological factors like maximum and minimum temperature, relative humidity, and rainfall. The significance of correlation coefficient was judged by 't' test as per standard statistical methods.

The data on disease intensity were plotted against meteorological parameters to predict the correlation between disease and environment. Regressions equations were also computed and scatter diagram were drawn.

Data on meteorological observations during the period was obtained from Meteorological Department, R.A.K. college of Agriculture, Sehore on temperature, relative humidity, (Table-1). These were plotted against various degrees of disease intensities. On the

basis of week to week observations, environmental conditions prevailed during the course of disease development were examined.

Table 1: Meteorological data during the crop season of 2013-14 at Sehore

Month	Standard Meteorological week No.	Dates	Temperature (°C)		Relative humidity (%)
			Max.	Min.	
November	45	4-10	29.1	16	62.15
	46	11-17	26.2	8	49.90
	47	18-24	27.3	7.2	55.32
	48	25-01	28.2	9.3	54.21
December	49	02-08	27.7	7.5	57.69
	50	09-15	25.9	7.4	47.40
	51	16-22	25.1	6.2	48.08
	52	23-29	24.1	10.1	58.11
January	1	30-05	25.7	12.1	61.21
	2	06-12	25.6	7.4	62.64
	3	13-19	25.6	7.9	59.28
	4	20-26	25.9	10	71.90
	5	27-02	24.8	6.1	72.32
February	6	03-09	29.4	9.1	65.56
	7	10-16	25.4	8.7	63.14
	8	17-23	29.6	4.5	38.41
	9	24-02	26.8	11.8	55.31
March	10	03-09	28.1	9.2	48.96
	11	10-16	33.6	15.2	41.23
	12	17-23	34.9	13.1	49.98
	13	24-30	36.2	17.5	51.20

Source: Meteorological observatory, R.A.K. College of Agriculture, Sehore (M.P.)

2.6 Management of disease by biological agent and chemicals

A field trial was conducted to find out more effective and suitable biological and chemical treatment for the control of dry root rot of chickpea and their growth enhancing ability

Design : Randomized block design
 Plot size : 4 x 1.8 m
 Row to row : 30 cm
 Seed / plot : 96 g

No. of rows	:	7
Plot to plot distance	:	60 cm
Replications	:	3
Treatments	:	10
Crop	:	Chickpea
Variety	:	KAK-2

Treatments

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed

T₂: Seed treatment by *B. subtilis* @ 4g/kg of seed

T₃: Seed treatment by *B. subtilis* @ 2g/kg foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing

T₄: Seed treatment by *B. subtilis* @ 4g/kg foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing

T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30, 45 and 60 days of sowing

T₆: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30, 45 and 60 days of sowing

T₇: Thirum + Carbendazim @ (2:1) (0.3%)

T₈: *Trichoderma* + Carboxin @ (4:1g)/kg

T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg

T₁₀: Untreated control

Observations:

(a) Effect on germination

Germination per cent of chickpea was recorded by counting the complete plant population of each plot after thirty days of sowing and percentage emergence was calculated on the basis of seeds sown/plot. The data were transformed angularly and analyzed.

(b) Incidence of dry root rot

Incidence of dry root rot were taken at regular interval of 7 days

(c) Shoot and root length

Shoot and root length of each plant were recorded in cm. The shoot and root length of plant were recorded after 30 and 60 days of sowing and mean of 10 plants was calculated.

(d) Fresh and dry weight of shoot

Fresh and dry weight of each plant was recorded in g. The fresh and dry weight of plant was recorded after 30 and 60 days of sowing and mean of 10 plants was calculated. The sample was weighed with the help of electronic balance.

(e) Fresh and dry weight of root

Fresh and dry weight of root was recorded in g. The fresh and dry weight of root was recorded after 30 and 60 days of sowing and mean of 10 plants was calculated. The sample was weighed with the help of electronic balance.

(f) Number of nodules, fresh and dry weight of nodules

Numbers of nodule, fresh and dry weight (mg) of each plant were recorded after 30 and 60 days of sowing and mean of 10 plants was calculated.

(g) Effect on yield components

The yield contributing characters like, number of pods, number of seeds per pod, seed yield per plant and seed yield per plot were recorded on 10 plants tagged in treatment of the experiment.

Statistical analysis

The data were subjected to statistical analysis after transformation. The data converted into percentage were transformed to angular values.

The difference between the two means was subjected to further testing by computing critical difference at 5% probability level.

(1) Standard error for treatment mean:

$$S. Em. \pm = \sqrt{Em.s / r}$$

(2) Critical difference:

$$C.D. = S. Em. \times \sqrt{2} \times t$$

Where,

Ems = Error means sum of square

r = Number of replication

t = 't' value at 5% probability levels

(3) Correlation coefficient:

Correlation and regression between percent disease index and meteorological parameters were worked out as per method suggested by Chandel (1999).

$$r(xy) = \frac{\text{COV.}(xy)}{\sqrt{\text{Var.}(x) \text{Var.}(y)}}$$

Where,

$r(xy)$ = correlation coefficient between character x and y.

$\text{COV}(xy)$ = covariance between character x and y.

$V(x)$ = variance of x character.

$V(y)$ = variance of y character.

Test of Significance of “r”:

$$t = \frac{r}{\sqrt{1-r^2}} \sqrt{n-2}$$

Where,

r = correlation coefficient

n = Number of observations

df = $(n-2)$

CHAPTER IV

RESULTS

1 Survey of chickpea crop for dry rot

Survey of chickpea crop was conducted in the second and third week of February 2014 at five tehsils of Sehore district in 28 locations. Percent disease index (PDI) was recorded and presented in Table-1.

Out of six villages of Sehore tehsil highest per cent disease index of *R. bataticola* was noticed in village Jatakheda (37.40%) followed by 35.60% in Mograram on KAK 2 variety (Table-1). However, PDI was lowest 22.30 and 17.90% in villages Naplakhedi and Sonda, respectively.

In Shyampur block out of six villages, three villages viz., Mungawali (34.30%), Shyampur (36.40%) and Khandwa (37.30%) showed disease incidence above 30%. In variety KAK 2 disease incidence was 17.40% in Sharkheda and 24.50% in Ravankheda. In variety Dollar 20% disease incidence was noticed in Lasudliya.

Out of six villages of Ichhawar tehsil, Palkhedi showed the highest incidence of *R. bataticola* (38.20%) in variety Kripa followed by Barkheda kurmi (34.10%), Bhaukhedi (33.40%), Dhabla mata (28.70%), Kheri hatesingh (23.30%). Lowest incidence was noticed on Desi (katila) in Siradi village (17.80%).

Five villages of Ashta tehsil viz., Hakimabad, Dhanana, Arniya, Khamkheda and Ashta were surveyed. Highest per cent disease index (PDI) was observed in Dhanana (39.30%) and Hakimabad (37.20%) village on Dollar and KAK 2 varieties, respectively.

In Nasrulaganj tehsil out of five villages, only one village Nayagaon (34.50%) showed disease incidence above 30%. Per cent disease incidence was lower than 20% in Ladkui (19.40%) and Koshmi (18.20%). Among the tehsils Ashta tehsil showed the highest incidence of disease (33.16%).

Table-1: Root rot incidence in chickpea fields in different blocks of Sehore district.

S.No.	Block	Village	Cultivar	Date of sowing	Date of observation	Mean PDI (%)
1	Sehore	Haiderganj	Dollar	20.11.2013	12.02.2014	17.90
2		Naplakhedi	Kripa	12.11.2013	12.02.2014	22.30
3		Aladakhedi	JG130	24.10.2013	12.02.2014	33.60
4		Lalakhedi	KAK 2	18.11.2013	12.02.2014	37.40
5		Hasnabad	Desi (Katila)	17.11.2013	12.02.2014	28.90
6		Mograram	KAK 2	20.11.2013	12.02.2014	35.60
Mean PDI (%)						29.28
7	Shyampur	Lasudliya	Dollar	17.11.2013	13.02.2014	20.90
8		Mungawali	Vishal	17.11.2013	13.02.2014	34.30
9		Shyampur	JG130	21.11.2013	13.02.2014	36.40
10		Sharkheda	KAK 2	20.10.2013	13.02.2014	17.40
11		Ravankheda	KAK 2	16.11.2013	14.02.2014	24.50
12		Khandwa	Vishal	18.11.2013	14.02.2014	37.30
Mean PDI (%)						28.45
13	Ichhawar	Barkheda kurmi	Desi (katila)	18.11.2013	15.02.2014	34.10
14		Dhablamata	Dollar	22.10.2013	15.02.2014	28.70
15		Siradi	Desi (katila)	23.10.2013	15.02.2014	17.80
16		Bhaukhedi	Dollar	20.11.2013	15.02.2014	33.40
17		Kheri	Kripa	19.11.2013	15.02.2014	23.30
18		Palkhedi	Kripa	20.11.2013	15.02.2014	38.20
Mean PDI (%)						29.25
19	Ashta	Arniya	Vishal	25.11.2013	14.02.2014	31.30
20		Khamkheda	Vishal	18.11.2013	14.02.2014	23.80
21		Ashta	KAK 2	17.11.2013	14.02.2014	34.20
22		Dhanana	Dollar	22.11.2013	16.02.2014	39.30
23		Hakimabad	Vishal	24.11.2013	16.02.2014	37.20
Mean PDI (%)						33.16
24	Nasrullaganj	Ladkui	JG130	19.11.2013	17.02.20.14	19.40
25		Badnagar	Vishal	21.11.2013	17.02.20.14	24.30
26		Nayagaon	Vishal	24.11.2013	17.02.20.14	34.50
27		Hallyakhedi	KAK 2	22.11.2013	17.02.20.14	18.20
28		Nasrullaganj	Jaki 9218	18.11.2013	17.02.20.14	21.50
Mean PDI (%)						23.58

4.2 Isolation and Identification of associated fungi

The incidence of dry root rot was mostly observed during flowering to podding stage of the crop. The first symptom of the disease is yellowing of the leaves within a day or two such leaves droop and while plant die off in the next two or three days. The leaves and stem of the affected plants were straw coloured. The tap root was dark and quite brittle in dry soil. The dark black sclerotia observed on and within host tissue of stem. Dry root rot affected plants were pulled bulked easily and showed blackening of roots (Plate-1). The tissues were weakened and break off easily. In advanced cases sclerotial bodies were seen scattered on the affected tissue. Isolation made from roots of infected chickpea samples yielded growth of *Rhizoctonia bataticola*. The *Rhizoctonia* isolates were purified by hyphal tip method, sub cultured and maintained on PDA slants for further studies as per the title of the thesis. The isolates were referred as RBS.

The sixteen isolates from roots of infected chickpea plants were identified as *R. bataticola* on the basis of morphological characters, viz., radial hyaline colonies which later became carbonaceous brown to black, mycelium septate and dark brown in colour, typical right angled branching of mycelium and production of sclerotia.

4.3 Pathogenicity tests

The pathogenicity of the *R. bataticola* (Taub) Butler [*Macrophomina phaseolina* (Tassi) Goid] isolated from the affected chickpea plants confirmed by multiplying the fungus and mixing it in sterile sand in pot. Symptoms of root rot appeared at 15-20 days in susceptible chickpea variety L-550. The affected plants exhibited sudden drying and root rot symptoms. The total final mortality was 95 per cent. Reisolation from the roots of the root rot affected plants consistently yielded *R. bataticola* sclerotial stage of the fungus *M. phaseolina*, thus confirming the pathogenic potential of the isolated fungus (Plate-2).

4.4 Variability studies

4.4.1 Cultural and Morphological variability

4.4.1.1 Mycelium characters

The mycelium was pale white in colour in the initial stage of the growth but later turned dark brown to black. Details of growth pattern and color of mycelium for all 16 isolates after 7th day of incubation are given in Table-2a and Plate-2.

Table-2a: Variability in cultural characteristics of *Rhizoctonia bataticola* isolate

S. No.	Isolates	Growth of mycelium	Colour of mycelium
1	RBS-1	Suppressed	Dark brown
2	RBS-2	Aerial	Whitish grey to black
3	RBS-3	Suppressed	Grey to black
4	RBS-4	Suppressed	Dark brown
5	RBS-5	Suppressed	Dark brown
6	RBS-6	Suppressed	Dark brown
7	RBS-7	Aerial	Whitish grey to black
8	RBS-8	Suppressed	Brown to black
9	RBS-9	Aerial	Whitish grey
10	RBS-10	Suppressed	Black
11	RBS-11	Suppressed	Grey to Black
12	RBS-12	Suppressed	Black
13	RBS-13	Suppressed	Dark brown
14	RBS-14	Suppressed	Grey to black
15	RBS-15	Suppressed	Whitish grey to black
16	RBS-16	Suppressed	Whitish grey to black

Based on the visual observation on *R. bataticola* isolates were divided into two groups, suppressed and aerial. Isolates RBS-1, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-10, RBS-11, RBS-12, RBS-13, RBS-14, RBS-15 and RBS-16 showed suppressed mycelium growth, whereas isolates RBS-2, RBS-7 and RBS-9 showed aerial growth.

Table-2b: Grouping of isolates of *R. bataticola* on the basis of mycelium colour and growth

Mycelial colour	Isolates	Mycelial growth	Isolates
Dark brown	RBS-1, RBS-4, RBS-5, RBS-6 and RBS-13	Aerial	RBS-2, RBS-7 and RBS-9
Whitish grey to black	RBS-2, RBS-7, RBS-15 and RBS-16	Suppressed	RBS-1, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-10, RBS-11, RBS-12, RBS-13, RBS-14, RBS-15 and RBS-16
Grey to black	RBS-3, RBS-11 and RBS-14		
Brown to black	RBS-8		
Whitish grey	RBS-9		
Black	RBS-10 and RBS-12		

Isolates were characterized into six groups on the basis of mycelium colour. Isolates RBS-1, RBS-4, RBS-5, RBS-6 and RBS-13 showed dark brown mycelium colour. Whitish grey to black colour of mycelium was observed in four isolates viz., RBS-2, RBS-7, RBS-15 and RBS-16. Isolates RBS-3, RBS-11 and RBS-14 showed grey to black coloured colonies whereas, isolate RBS-8 showed brown to black colour. Only one isolate RBS-9 showed whitish grey colour. Isolates RBS-10 and RBS-12 showed black coloured colonies (Table-2b and Plate-3).

Growth per day

Studies on growth rate of isolates were conducted to know the fast growing, medium growing and slow growing isolates at different intervals on P.D.A. Observations on growth rate of different isolates were made after each day up to 5th day of incubation (Table-3).

On the basis of average growth per day the data denotes that the slow growth (9.5-12.5 mm growth/day) has been obtained in isolates RBS-4, RBS-8, RBS-13 and RBS-15 while isolates RBS-1, RBS-3, RBS-5, RBS-6, RBS-7, RBS-10, RBS-12, RBS-14 and RBS-16 showed medium growth (12.6-15.5

mm growth/day). Rest of the isolates RBS-2, RBS-9 and RBS-11 attained the fast growth (15.6-18.0 mm growth/day).

Table-3: Growth rate of various isolates of *R. bataticola* on potato dextrose agar medium at different intervals

Isolates No.	Growth rate (mm / day)					Average growth per day (mm)
	1	2	3	4	5	
RBS-1	15	28	45	60	77	15.40
RBS-2	24	38	53	69	90	18.00
RBS-3	8	22	35	54	71	14.20
RBS-4	0	8	20	38	59	11.80
RBS-5	9	19	32	46	67	13.40
RBS-6	12	25	39	57	78	15.60
RBS-7	11	26	42	58	72	14.40
RBS-8	6	13	28	44	62	12.40
RBS-9	22	34	52	65	84	16.80
RBS-10	5	14	29	45	66	13.20
RBS-11	23	37	56	75	90	18.00
RBS-12	13	27	45	58	77	15.40
RBS-13	0	7	17	35	58	11.60
RBS-14	18	30	46	61	75	15.00
RBS-15	0	6	18	37	61	12.20
RBS-16	17	29	46	59	76	15.20
SEm±						0.14
C.D. at 5%						0.39

Table-4: Grouping of isolates of *R. bataticola* on the basis of growth rate

Characters	Isolates	Total
Slow growth (9.5-12.5 mm)	RBS-4, RBS-8, RBS-13 and RBS-15	4
Medium growth (12.6-15.5 mm)	RBS-1, RBS-3, RBS-5, RBS-6, RBS-7, RBS-10, RBS-12, RBS-14 and RBS-16	9
Fast growth (15.6-18.0 mm)	RBS-2, RBS-9 and RBS-11	3
Total		16

4.4.1.2 Sclerotia characters

Results (Table-5(a&b)&Plate-4) revealed that all the sixteen representative isolate of *R. bataticola* exhibited a wide range of variability in respect of size of sclerotia. Average size of sclerotia of the test isolates was ranged from 36.16 x 21.96 µm to 72.84 x 47.06 µm. Maximum sclerotia size recorded in isolate of RBS-16 (72.84x47.06 µm) followed by RBS-12, RBS-3 and RBS-14.

Isolate RBS-1 and RBS-4 produced medium size sclerotia whereas, Isolates RBS-9, RBS-5, RBS-6, RBS-13, RBS-11, RBS-8, RBS-10, RBS-2 and RBS-15 produced small size sclerotia compared to other isolates. Minimum sclerotia size (36.16x21.96µm) was recorded in isolate of RBS-7.

Slight variation was observed in sclerotia colour of the isolates. Isolates RBS-1, RBS-7, RBS-14 and RBS-15 produced dark brown coloured sclerotia whereas, isolates RBS-2, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-9, RBS-10, RBS-11, RBS-12, RBS-13 and RBS-16 produced black coloured sclerotia (Table-5a and Plate-4).

Table-5a: Variability in morphological characters of *Rhizoctonia bataticola* sclerotia

S. No.	Isolates	Sclerotia size(µm)	Sclerotia colour
1	RBS-1	64.14 x 50.00	Dark brown
2	RBS-2	45.40 x 33.58	Black
3	RBS-3	69.14 x 41.30	Black
4	RBS-4	63.98 x 46.22	Black
5	RBS-5	53.12 x 41.72	Black
6	RBS-6	53.20 x 35.40	Black
7	RBS-7	36.16 x 21.96	Dark brown
8	RBS-8	47.80 x 35.96	Black
9	RBS-9	55.98 x 30.80	Black
10	RBS-10	46.56 x 37.30	Black
11	RBS-11	50.70 x 35.30	Black
12	RBS-12	69.50 x 51.24	Black
13	RBS-13	51.50 x 34.50	Black
14	RBS-14	69.92 x 39.32	Dark brown
15	RBS-15	41.86 x 31.90	Dark brown
16	RBS-16	72.84 x 47.06	Black

Number of sclerotia per microscopic field were also observed for each isolate and grouped into, three groups viz; minimum (5-20), medium (21-40),

maximum (40<) (Table-5b). The number of sclerotia varied from 5.80 to 53.00. Out of 16 isolates, 8 isolates were observed under minimum number of sclerotia, 6 isolates were grouped under medium number of sclerotia and 2 isolates were categorized under maximum number of sclerotia. Among all the isolates, the isolate RBC-12 had maximum number of sclerotia 53.0 while the minimum number of sclerotia 6.7 were observed in isolate RBC-10.

Table-5b: Grouping of isolates of *R. bataticola* on the basis of sclerotia characters

Sclerotia colour	Isolates	Number of sclerotia / microscopic field	Isolates
Dark brown	RBS-1, RBS-7, RBS-14 and RBS-15	Minimum (5-20)	RBC-3, RBC-4, RBC-5, RBC-6, RBC-9, RBC-11, RBC-15, RBC-16
Black	RBS-2, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-9, RBS-10, RBS-11, RBS-12, RBS-13 and RBS-16	Medium (21-40)	RBC-1, RBC-7, RBC-8, RBC-10, RBC-12, RBC-13,
		Maximum (40<)	RBC-2, RBC-14

4.4.2 Pathogenic variability

The data presented in table-6 and Fig.-1 show that the mortality percentage was ranged between 0.00-100.00%. The maximum mortality (100.00%) was observed in isolate RBS-1 with JG-62 and in isolate RBS-2 with JG-11 whereas, the minimum mortality (0.00%) was founded in isolate RBS-1 with BGD-112, RBS-5 with RVG-205, RBS-8 with JKG-1 and RBS-10 with RVG-205 and BGD-112. The root index percentage varied from 26.66 to 91.66. Minimum 26.66 per cent was recorded in I-9 followed by 27.50 per cent in 19 respectively. Highest root index 91.66 and 83.30 was observed in I-10 isolate and I-2 isolate. Mortality percent per unit ranged between 0.83 to 1.59 out of 16 isolates screened. None of them was categorized below seven disease rating.

Table-6: Disease incidence in mortality percentage of different varieties of chickpea due to *Rhizoctonia bataticola*

Varieties	Isolates of <i>Rhizoctonia bataticola</i>																Mean	R	MR	S	HS
	RBS-1	RBS-2	RBS-3	RBS-4	RBS-5	RBS-6	RBS-7	RBS-8	RBS-9	RBS-10	RBS-11	RBS-12	RBS-13	RBS-14	RBS-15	RBS-16					
RVG 202	20.00 (26.07)	36.67 (37.14)	13.33 (21.14)	23.33 (18.93)	23.33 (28.78)	46.67 (43.08)	50.00 (44.71)	20.00 (26.07)	20.00 (26.07)	30.00 (32.30)	46.67 (43.08)	33.33 (34.22)	26.67 (30.29)	6.67 (8.86)	16.67 (15.00)	20.00 (26.07)	27.08 (28.86)	1	10	5	0
JG-16	73.33 (59.00)	96.67 (83.86)	83.33 (66.64)	86.67 (68.86)	90.00 (75.00)	83.33 (66.14)	93.33 (81.14)	96.67 (83.86)	96.67 (83.86)	96.67 (83.86)	96.67 (83.86)	96.67 (83.86)	93.33 (77.71)	10.00 (11.07)	33.33 (35.22)	63.33 (53.15)	80.63 (68.57)	1	0	1	14
RVG 203	16.67 (23.86)	13.33 (21.14)	10.00 (15.00)	20.00 (22.14)	0.00 (0.00)	3.33 (6.14)	0.00 (0.00)	10.00 (15.00)	0.00 (0.00)	23.33 (24.15)	13.33 (17.71)	3.33 (6.14)	13.33 (21.14)	6.67 (12.29)	33.33 (30.00)	6.67 (12.29)	10.83 (14.19)	9	6	1	0
BGD-112	3.33 (6.14)	23.33 (24.15)	0.00 (0.00)	16.67 (23.86)	0.00 (0.00)	10.00 (15.00)	16.67 (23.86)	3.33 (6.14)	20.00 (26.07)	26.67 (30.29)	16.67 (19.93)	16.67 (19.93)	36.67 (37.22)	0.00 (0.00)	6.67 (8.86)	6.67 (12.29)	12.71 (15.86)	8	7	1	0
JKG-1	3.33 (6.14)	10.00 (15.00)	10.00 (15.00)	26.67 (30.29)	13.33 (21.14)	0.00 (0.00)	3.33 (6.14)	0.00 (0.00)	3.33 (6.14)	10.00 (15.00)	23.33 (28.28)	13.33 (17.21)	30.00 (33.00)	3.33 (6.14)	30.00 (27.29)	23.33 (28.28)	12.71 (15.94)	9	7	0	0
L-550	36.67 (31.92)	46.67 (42.99)	46.67 (42.00)	86.67 (72.79)	20.00 (22.14)	50.00 (44.71)	16.67 (19.93)	16.67 (19.93)	46.67 (43.08)	80.00 (63.93)	63.33 (53.86)	16.67 (23.86)	43.33 (40.78)	10.00 (15.00)	13.33 (17.71)	6.67 (12.29)	37.50 (35.43)	2	5	6	3
Annegiri	63.33 (53.36)	76.67 (61.92)	60.00 (50.85)	83.33 (66.14)	70.00 (57.00)	86.67 (72.29)	63.33 (53.15)	63.33 (53.07)	36.67 (37.22)	46.67 (43.08)	76.67 (61.92)	56.67 (48.85)	56.67 (48.93)	16.67 (15.00)	30.00 (28.08)	50.00 (44.92)	58.54 (49.74)	0	2	3	11
JSC-37	83.33 (66.64)	60.00 (51.14)	60.00 (50.94)	60.00 (50.85)	73.33 (59.00)	66.67 (55.36)	63.33 (53.15)	50.00 (45.00)	63.33 (53.15)	66.67 (54.78)	73.33 (60.79)	73.33 (59.21)	63.33 (52.78)	20.00 (16.92)	33.33 (35.22)	36.67 (37.22)	59.17 (50.14)	0	1	3	12
JG-62	63.33 (53.15)	96.67 (83.86)	100.00 (90.00)	93.33 (77.71)	83.33 (66.64)	86.67 (72.79)	70.00 (57.00)	93.33 (77.71)	86.67 (68.86)	90.00 (71.57)	93.33 (77.71)	83.33 (66.14)	96.67 (83.86)	13.33 (17.21)	50.00 (44.92)	43.33 (41.07)	77.71 (65.64)	0	1	2	13
JG-11	83.33 (66.14)	76.67 (65.85)	83.33 (70.78)	90.00 (75.00)	70.00 (57.29)	76.67 (66.93)	83.33 (70.07)	66.67 (55.07)	76.67 (66.14)	76.67 (66.14)	100.00 (90.00)	83.33 (75.00)	80.00 (68.07)	30.00 (23.86)	50.00 (45.29)	40.00 (38.86)	72.92 (62.53)	0	1	2	13
Mean	44.67 (39.24)	53.67 (48.71)	46.67 (42.24)	58.67 (50.66)	44.33 (38.70)	51.00 (44.24)	46.00 (40.92)	42.00 (38.19)	45.00 (41.06)	54.67 (48.51)	60.33 (53.71)	47.67 (43.44)	54.00 (49.38)	11.67 (12.64)	29.67 (28.76)	29.67 (30.64)					

Ranged **R**= 0-10 **MR**= 11-30, **S**= 31-50, **HS**= 51 above

A critical look at the results revealed that all the isolates showed various reactions with different varieties. The data indicate that all the isolates showed more than 40 per cent mortality excepted RBS-4, RBS-13 and RBS-15. Out of sixteen isolates, only six isolates RBS-14 (51.00%), RBS-12 (53.67%), RBS-9 (54.00%), RBS-7 (54.67%), RBS-11 (58.67%) and RBS-2 (60.33%) caused >50 per cent dry root rot incidence and the remaining seven isolates RBS-8, RBS-10, RBS-6, RBS-5, RBS-16, RBS-1 and RBS-3 (42.00, 44.33, 44.67, 45.00, 46.00, 46.67 and 47.67%, respectively) caused 40-50 per cent dry root rot incidence (Table-6).

The data presented in table-6 clearly indicated that out of 10 varieties tested against 16 isolate of *R. bataticola* none of the entries were resistant (0-10% mortality). Only three varieties, namely RVG-203 (10.83%), BGD -112 (12.71%) and JKG-1 (12.71%) had shown moderately resistant reaction. Five varieties Annegiri, L-550, JG-11, JG-62 and JG-16 (58.54, 59.17, 72.92, 77.71 and 80.63%, respectively) exhibited highly susceptible reaction (more than 50% mortality) against 16 isolates of *R. bataticola*.

4.5 Effect of environmental factors on incidence of dry root rot of chickpea

The development of dry root rot was studied in relation to various climatic parameters. The dry root rot incidence was recorded every week and correlated with mean maximum and minimum temperature, relative humidity, rainfall (mm) and number of rainy days of the corresponding meteorological weeks. Correlation coefficient (r) values were computed for each parameter and given in table-7a. To quantify the effect of various parameters the linear regression equation was also worked out.

Table-7a: Correlation coefficient (r) values between intensity of dry root rot of chickpea and various weather parameters

Weather parameter	Correlation coefficient (r)	Test of significance (t)
Maximum temperature	0.80*	5.26
Minimum temperature	0.66*	3.51
Relative humidity	-0.38	-1.66

*Significant at 5% level of significance

Table-7b: Regression equation between intensity of dry root rot of chickpea and various weather parameters

Weather parameter	Regression equation
Maximum temperature	$Y = 0.141x + 24.86$
Minimum temperature	$Y = 0.109x + 7.14$
Relative humidity	$Y = -0.190x + 59.85$

4.5.1 Maximum temperature

Maximum temperature was found to have a significant positive correlation with dry root rot of chickpea ($r=0.80$). The regression equation showed that dry root rot intensity increased by 0.141% with the increase of 1°C maximum temperature (Table-7a, 7b and Fig.-2).

4.5.2 Minimum temperature

Minimum temperature was found to have a significant positive correlation with dry root rot of chickpea ($r=0.66$). The regression equation showed that dry root rot intensity decreased by 0.109% with increase of 1°C minimum temperature (Table-7b and Fig.-3).

4.5.3 Relative humidity

Relative humidity was found to have a significant negative correlation with dry root rot intensity ($r=-0.38$). The regression equation indicated that with the every 1 per cent increase in relative humidity the disease intensity decreased by 0.190 per cent (Table-7b and Fig.-4).

4.6. Evaluation of biological agent and chemicals for management dry root rot

Incidence of dry root rot caused by *R. bataticola* was significantly less in all the treatments as compared to control (Table-8 and Fig.-5). (T8) *Trichoderma*+Carboxin (14.58%) was the best treatment in controlling root rot of chickpea followed by (T9) *Trichoderma*+Amm. Molybdate (16.37%), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (17.49%) and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (18.32%) as compared to other treatment. In control, where no treatment was done the incidence of root rot was 68.22 percent.

Table-8: Incidence of dry root rot of chickpea in different *Bacillus subtilis* treatments

Treatment	Incidence of dry root rot (%)
T1	27.47 (31.61)**
T2	23.26 (28.84) **
T3	18.32 (25.34) **
T4	17.49 (24.72) **
T5	21.79 (27.82) **
T6	19.82 (26.44) **
T7	24.07 (29.38) **
T8	14.58 (22.45) **
T9	16.37 (23.86) **
Control	68.22 (55.69) **
SEm±	0.25
C.D. at 5%	0.53

*mean of three replications

** Data in parentheses are arcsine transformed values

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, **T₂**: Seed treatment by *B. subtilis* @ 4g/kg of seed, **T₃**: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, **T₄**: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, **T₅**: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, **T₆**: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, **T₇**: Thirum + Carbendazim @ (2:1) (0.3%), **T₈**: *Trichoderma* + Carboxin @ (4:1g)/kg, **T₉**: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, **T₁₀**: Untreated control.

4.7 Evaluation of biological agent and chemicals for growth promoting ability (GPA) of chickpea

4.7.1 Effect of plant vigour

Significant differences occurred with respect to plant vigour in different treatments. The plant vigour ranged between 2306 to 5555.

Table-9: Chickpea plant vigour in different *B. subtilis* treatments

Treatment	Plant vigour
T1	3812
T2	4394
T3	5031
T4	5121
T5	4636
T6	4944
T7	3770
T8	5555
T9	5182
Control	2306
SEm±	135.28
C.D. at 5%	284.23

*mean of three replications

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, **T₂**: Seed treatment by *B. subtilis* @ 4g/kg of seed, **T₃**: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, **T₄**: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, **T₅**: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, **T₆**: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, **T₇**: Thirum + Carbendazim @ (2:1) (0.3%), **T₈**: *Trichoderma* + Carboxin @ (4:1g)/kg, **T₉**: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, **T₁₀**: Untreated control.

Significantly highest plant vigour was recorded in (T8) *Trichoderma*+Carboxin (5555) followed by (T9) *Trichoderma*+Amm. Molybdate (5182), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (5121) and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (5031) as compared to other treatment and control (Table-9 and Fig.-6).

4.7.2 Fresh and dry weight of chickpea shoot

The data presented in table -10 shows the effect of seed dressing with *Trichoderma* strains on shoot fresh and dry weight of chickpea.

Data in the table-10 and Fig.-7 shows that all the treatments significantly increased fresh shoot weight after 30 and 60 days after sowing as compared to control. Significantly highest fresh shoot weight was observed in (T8) *Trichoderma*+Carboxin (27.07g) at 30 days after sowing followed by (T9) *Trichoderma*+Amm. Molybdate (26.10 g), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (25.90 g) and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (25.80g) as compared to other treatments and control. Significantly after 60 days of sowing highest shoot fresh weight was noticed in (T8) *Trichoderma*+Carboxin (64.40 g) followed by (T9) *Trichoderma*+Amm. Molybdate (62.07 g), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (60.30 g) and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (60.27 g).

Table-10: Fresh and dry weight of shoot of chickpea in different treatments

Treatment	Fresh weight (g)		Dry weight (g)	
	30 DAS	60 DAS	30 DAS	60 DAS
T1	23.97	53.67	7.10	16.67
T2	24.77	54.07	8.40	16.93
T3	25.80	60.27	10.57	21.10
T4	25.90	60.30	10.70	21.27
T5	23.57	59.40	10.07	20.53
T6	25.56	60.23	10.30	20.60
T7	24.50	54.97	6.23	15.00
T8	27.07	64.40	11.30	22.17
T9	26.10	62.07	11.00	21.67
Control	20.20	36.80	3.87	13.00
SEm±	0.36	0.46	0.26	0.65
C.D. at 5%	0.76	0.97	0.54	1.36

*Average of three replications.

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, T₂: Seed treatment by *B. subtilis* @ 4g/kg of seed, T₃: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, T₄: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B.*

subtilis @ 250 g/ha after 30 and 45 days of sowing , T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, T₆: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, T₇: Thirum + Carbendazim @ (2:1) (0.3%), T₈: *Trichoderma* + Carboxin @ (4:1g)/kg, T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, T₁₀: Untreated control.

All treatments significantly increased the shoot dry weight as compared to control after 30 and 60 days after sowing. Significantly highest shoot dry weight/plant (11.30g) was obtained in T₈ (*Trichoderma*+Carboxin) which was at par with T₉ (*Trichoderma*+Amm. Molybdate) followed by (T₄) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (10.70 g) and (T₃) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (10.57 g) after 30 days of sowing. After 60 days of sowing, highest shoot dry weight per plant was obtained in (T₈) *Trichoderma*+Carboxin (22.17 g) and was at par with (T₉) *Trichoderma*+Amm. Molybdate (21.67 g), (T₄) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (21.27g) and (T₃) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (21.10g) (Table-10 and Fig.-8).

4.6.4 Root length, fresh and dry weight of root of chickpea

The root length varied significantly among different treatment (Table-11 and Fig.-9). More root length was noticed in all the treatments except in T₇ (Thirum+Carbendazim @ 2:1) after 30 and 60 days of sowing. The highest root length (13.00cm) after 30 days was recorded in (T₈) *Trichoderma*+Carboxin which was at par with (T₉) *Trichoderma*+Amm. Molybdate (12.67 cm), (T₄) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (11.67cm) and (T₃) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (11.33cm) followed by (T₆) Soil + foliar drench of *B. subtilis* @ 500g/ha after

30,45 and 60 days of sowing (11.23cm) and (T5) Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing (10.98cm).

After 60 days of sowing highest root length was obtained in (T8) *Trichoderma*+Carboxin (19.33 cm) which was at par with (T9) *Trichoderma*+Amm. Molybdate (18.67) and (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (18.33 cm) followed by (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (17.33 cm), (T6) Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing (17.00) and (T2) Seed treatment by *Bacillus subtilis* @ 4g/kg of seed (16.67 cm).

Table-11: Root length, fresh and dry weight of chickpea in different treatments

Treatment	Root length (cm)		Fresh wt (g)		Dry wt (g)	
	30 DAS	60 DAS	30 DAS	60 DAS	30 DAS	60 DAS
T1	9.33	12.33	8.43	19.37	3.67	8.10
T2	9.67	16.67	9.73	19.93	4.78	12.07
T3	11.33	17.33	11.87	22.37	5.40	12.30
T4	11.67	18.33	12.07	22.83	5.33	12.33
T5	10.98	14.00	11.67	21.80	4.70	12.07
T6	11.23	17.00	11.80	22.33	5.33	12.28
T7	9.00	11.00	9.67	17.27	3.57	7.53
T8	13.00	19.33	13.40	24.53	6.10	12.87
T9	12.67	18.67	12.93	24.13	5.40	12.43
Control	7.33	9.33	6.73	14.70	2.23	6.60
SEm±	0.81	0.83	0.26	0.46	0.20	0.17
C.D. at 5%	1.71	1.74	0.56	0.96	0.43	0.37

*Average of three replications.

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, T₂: Seed treatment by *B. subtilis* @ 4g/kg of seed, T₃: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, T₄: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, T₆: Soil + foliar

drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, T₇: Thirum + Carbendazim @ (2:1) (0.3%), T₈: *Trichoderma* + Carboxin @ (4:1g)/kg, T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, T₁₀: Untreated control.

Data in the table-11 shows that all the treatments significantly increased the root fresh and dry weight after 30 and 60 days after sowing as compared to control. Treatment (T8) *Trichoderma*+Carboxin significantly increase the fresh root weight (13.40 g) at 30 days after sowing which was at par with (T9) *Trichoderma*+Amm. Molybdate (12.93 g) as compared to other treatments and control. Similarly after 60 days of sowing significantly highest root fresh weight was observed in (T8) *Trichoderma*+Carboxin (24.53 g) at par with (T9) *Trichoderma*+Amm. Molybdate (24.13 g) as compared to control (Fig.-10).

Treatments significantly increased the root dry weight after 30 and 60 days of sowing as compared to control (Table-11 and Fig.-11). Significantly highest root dry weight/plant was obtained in (T8) *Trichoderma*+Carboxin (6.10 g and 12.87 g) which as compared to other treatments and control (2.23 g and 6.60 g) after 30 and 60 days of sowing followed by (T9) *Trichoderma*+Amm. Molybdate and (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing followed by (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, (T6) Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing and (T5) Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing.

4.6.5 Number of nodules and nodules fresh and dry weight

The data on effect of treatments on number of nodule per plant, nodules fresh and dry weight/ plant was presented in table-12. Significantly increase number of nodules/ plant was found in all the treatments after 30 and 60 days after sowing as compared to control. Highest number of nodules were observed in treatment T8 (*Trichoderma*+Carboxin) (30.33), which was at par with treatments T9 (*Trichoderma*+Amm. Molybdate) (28.67) and T4 (Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250

g/ha after 30 and 45 days of sowing) (28.33) as compared to control (12.67) after 30 days of sowing.

Table-12: Number of nodules and nodules fresh and dry weight of chickpea in different treatments

Treatment	No. of nodules*		Fresh weight (mg)*		Dry weight (mg)*	
	30 DAS	60 DAS	30 DAS	60 DAS	30 DAS	60 DAS
T1	23.33	46.67	28.00	123.33	7.03	42.43
T2	24.67	48.00	29.67	127.33	8.00	45.80
T3	28.00	52.33	34.67	150.00	11.27	51.77
T4	28.33	52.67	34.67	151.33	11.60	52.07
T5	26.67	51.67	33.00	147.62	10.20	49.87
T6	27.00	52.00	33.33	149.67	11.20	50.27
T7	19.67	43.67	25.33	122.00	6.67	41.67
T8	30.33	54.00	43.33	154.33	13.97	53.90
T9	28.67	53.00	36.33	151.67	11.83	52.43
Control	12.67	35.00	19.00	99.67	3.93	28.20
SEm±	0.95	2.32	2.06	3.59	0.20	0.93
C.D. at 5%	2.00	4.88	4.32	7.54	0.41	1.95

*Average of tree replications.

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, **T₂**: Seed treatment by *B. subtilis* @ 4g/kg of seed, **T₃**: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, **T₄**: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, **T₅**: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, **T₆**: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, **T₇**: Thirum + Carbendazim @ (2:1) (0.3%), **T₈**: *Trichoderma* + Carboxin @ (4:1g)/kg, **T₉**: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, **T₁₀**: Untreated control.

After 60 days of sowing, increase in nodule number/plant was noticed in treated treatments. The highest number of nodules 54.00 was obtained in

(T8) *Trichoderma*+Carboxin and was at par with all treatments excepted Seed treatment by (T1) *Bacillus subtilis* seed treatment (2g/kg of seed) by (T2) *Bacillus subtilis* seed treatment (4g/kg of seed) and (T7) Thirum+Carbendazim (0.3%) as compared to control (35.00).

Table-12 shows that all the treatments significantly increase the nodules fresh weight after 30 and 60 days of sowing as compared to control. Highest nodules fresh weight was obtained in (T8) *Trichoderma*+Carboxin (43.33 mg) at par with all the treated treatments excepted (T1) *Bacillus subtilis* seed treatment (2g/kg of seed) and (T7) Thirum+Carbendazim (0.3%) as compared to control.

After 60 days of sowing, significantly highest nodules fresh weight was obtained at (T8) *Trichoderma*+Carboxin (154.33 mg) and was at par with (T9) *Trichoderma*+Amm. Molybdate (151.67 mg), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (151.33 mg), (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (150.00 mg), (T6) Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing (149.67mg) and (T5) Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing (147.62mg) compared to control.

The nodules dry weight varied among different treatments (Table-12). The highest nodules dry weight (13.97 mg) was obtained in (T8) *Trichoderma*+Carboxin which was at par with (T9) *Trichoderma* + Amm. Molybdate (28.67) and (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (28.33) as compared to other treatments and control after 30 days of sowing. At 60 DAS maximum nodules dry weight was obtained in (T8) *Trichoderma*+Carboxin (53.90 mg) and was at par with (T9) *Trichoderma* + Amm. Molybdate (52.43 mg) as rest of treatments and compared to control.

4.7.0 Post harvest observation

4.7.1 Number of pods per plant

The data on effect of treatment with *Bacillus subtilis*, *Trichoderma* strain and chemical combinations and chemicals on number of pods per plant

of chickpea are presented in table-13 and Fig.-12. The average number of pods ranged between 32.11 to 49.67. All the seed treatments with *Trichoderma* strains significantly increased the number of pods/plant as compare to control. Significantly maximum average numbers of pods were recorded in (T8) *Trichoderma*+Carboxin (49.67) at par with (T9) *Trichoderma* + Amm. Molybdate (48.99) and (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (48.00) as compared to control.

Table-13: Yield of chickpea in different treatments under field condition

Treatment	Pod/ plant*	Seed/ plant*	Seed yield/ plant (g)*	Yield kg/ha*
T1	35.67	46.37	6.09	1490.74
T2	36.50	47.45	6.30	1534.72
T3	46.67	60.67	10.94	2033.80
T4	48.00	62.40	12.08	2103.24
T5	42.25	54.93	7.55	1936.58
T6	43.58	56.65	8.55	1964.35
T7	34.90	45.37	5.80	1326.85
T8	49.67	64.57	13.25	2143.52
T9	48.99	63.69	12.48	2125.00
Control	32.11	41.74	4.92	1151.39
SEm±	1.28	1.09	0.68	52.83
C.D. at 5%	2.69	2.29	1.43	110.99

*Average of tree replications.

Here,

T₁: Seed treatment by *Bacillus subtilis* @ 2g/kg of seed, T₂: Seed treatment by *B. subtilis* @ 4g/kg of seed, T₃: Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, T₄: Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, T₆: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, T₇: Thirum + Carbendazim @ (2:1) (0.3%), T₈: *Trichoderma* + Carboxin @ (4:1g)/kg, T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, T₁₀: Untreated control.

4.7.2 Number of seeds per plant

The data on effect of treatment with *Bacillus subtilis* as seed, soil and foliar application and seed dressing with *Trichoderma* strain and chemicals on number of seed per plant were presented in table-13 and Fig.-12. Significant increase number of seed/ plant was found in (T8) *Trichoderma*+Carboxin (64.57) and (T9) *Trichoderma* + Amm. Molybdate (63.69) treated plants as compared to control (41.74).

4.7.3 Effect on yield/ha

The yield/ha varied among different treatment (Table-13 and Fig.-13). The highest yield/ha was obtained 2143.52 kg in (T8) *Trichoderma*+Carboxin which was at par with (T9) *Trichoderma*+Amm. Molybdate (2125.00 kg), (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (2103.24 kg) and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing (2033.80 kg) as compared to rest of treatments and control (1151.39 kg).



Plate-1a: View of chickpea field affected by dry root rot caused by *Rhizoctonia bataticola*



Plate-1b: Root affected from Dry Root Rot



(a) Less pathogenic



(b) Pathogenic



(c) High Pathogenic



(d) Control

Plate 2 Pathogenicity of dry root rot by pot inoculation methods

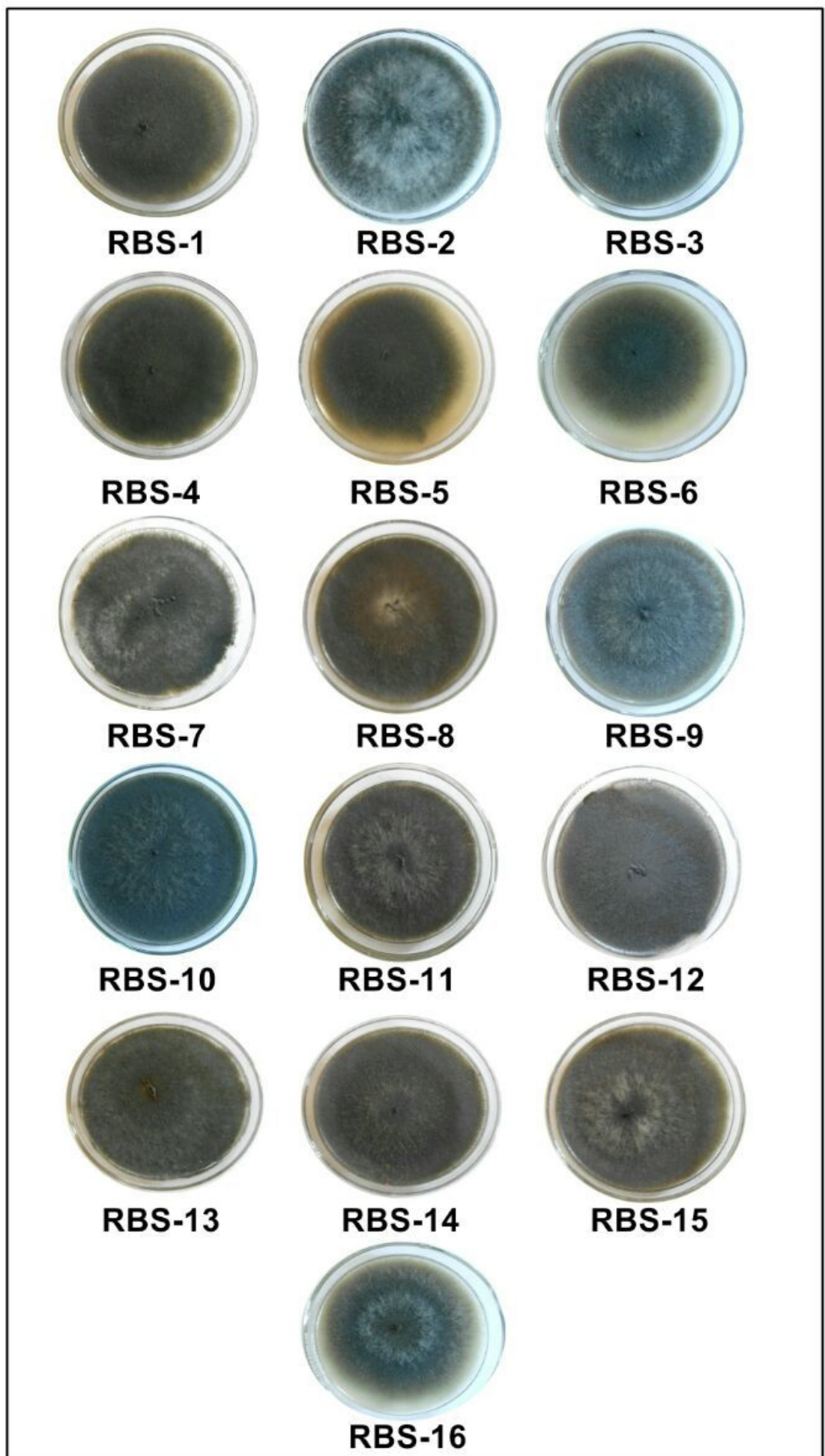


Plate-3: Variability in cultural characteristics of *Rhizoctonia bataticola* isolate

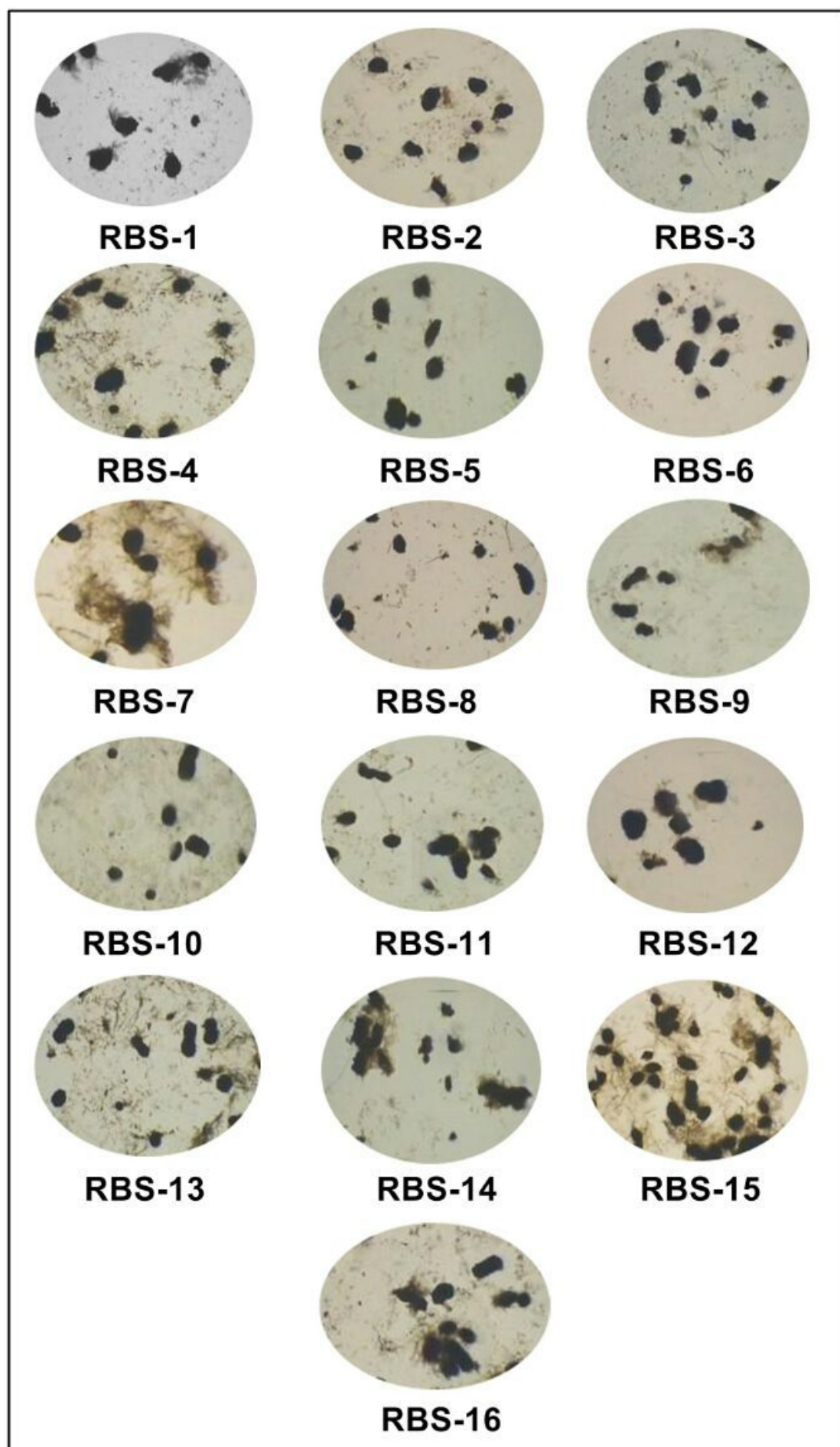


Plate-4: Variability in morphological characteristics of *Rhizoctonia bataticola* isolate

Fig-2: Regression line between disease incidence and maximum temperature (°C)

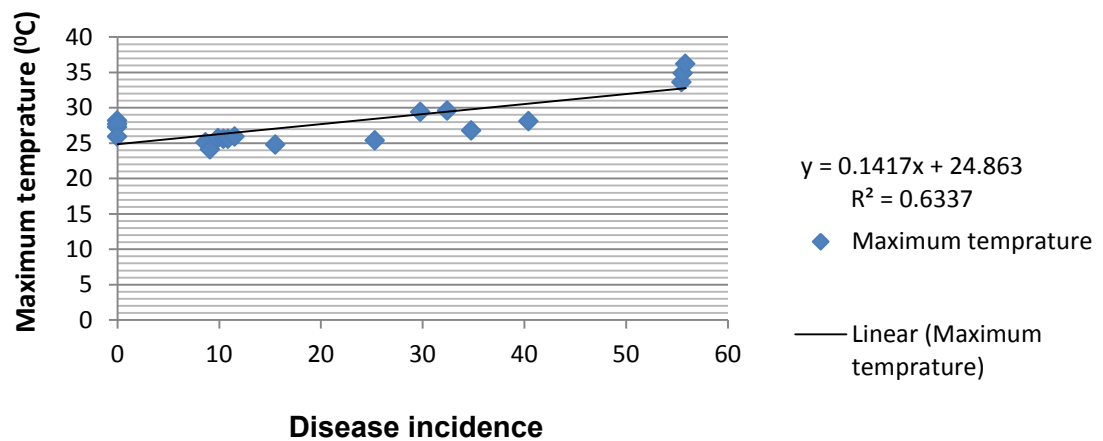


Fig-3: Regression line between disease incidence and minimum temperature (°C)

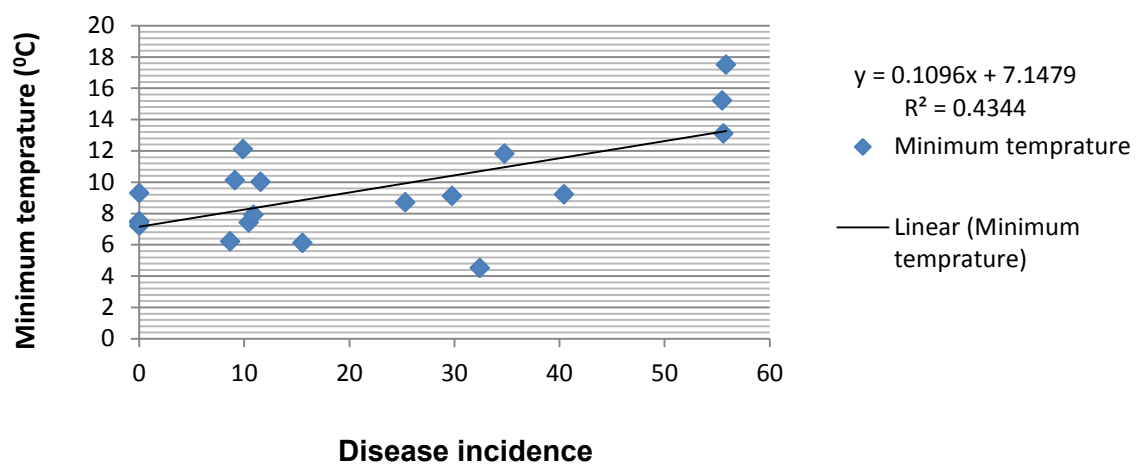
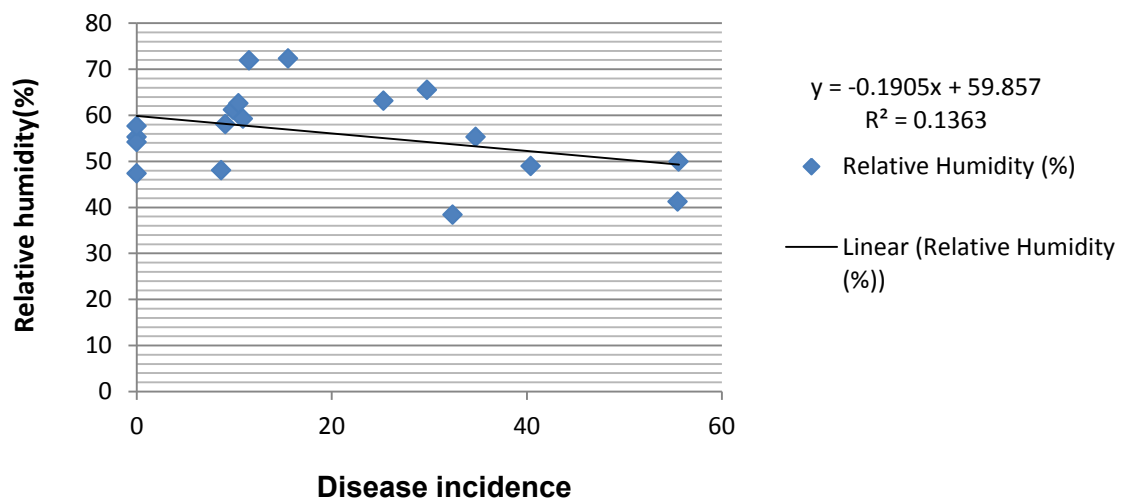
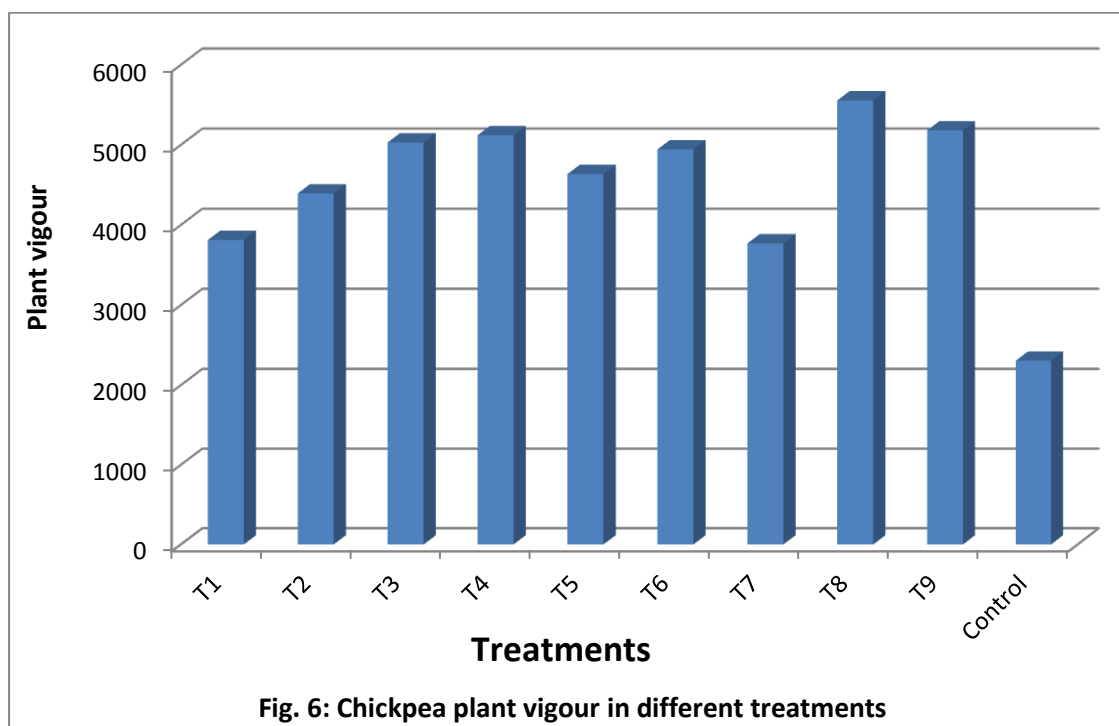
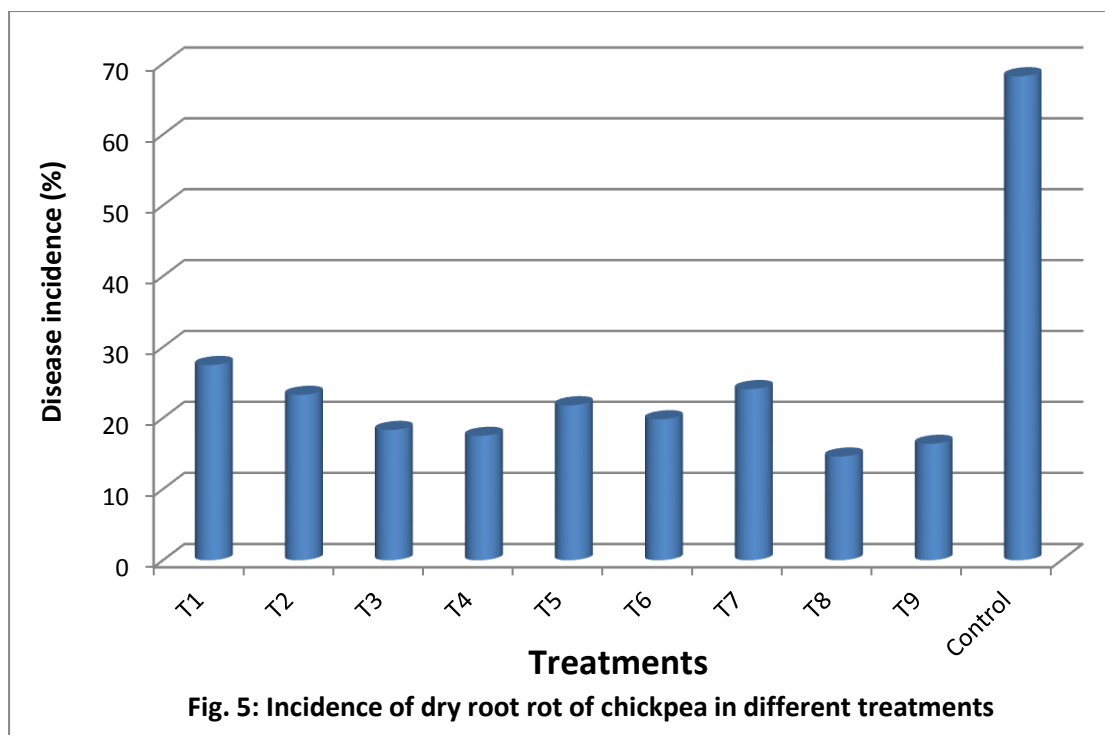
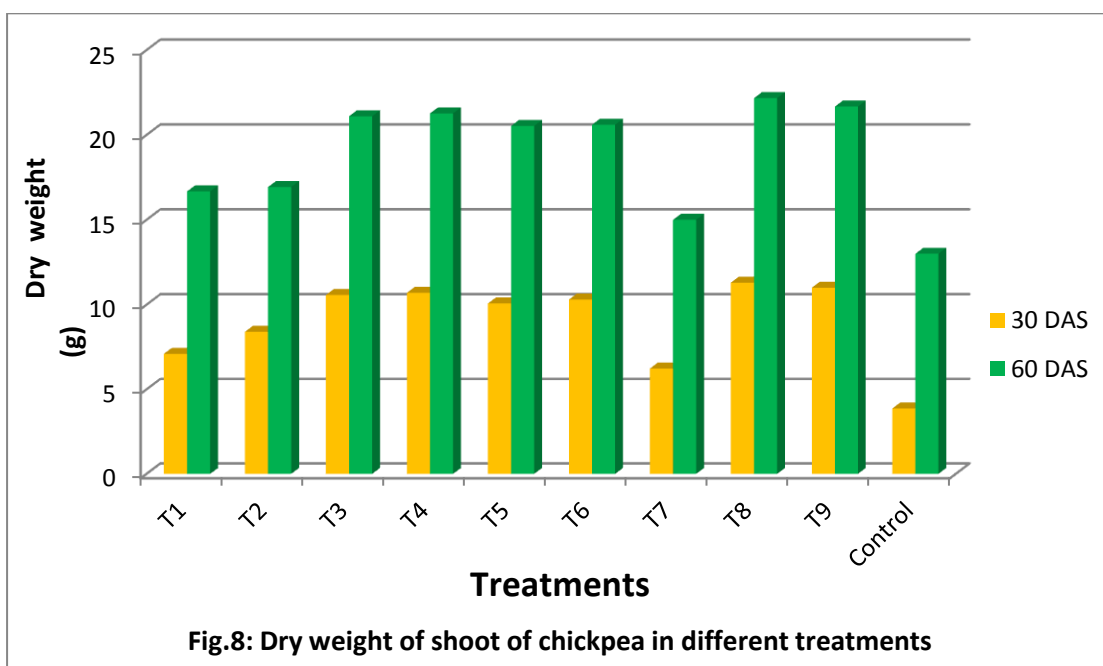
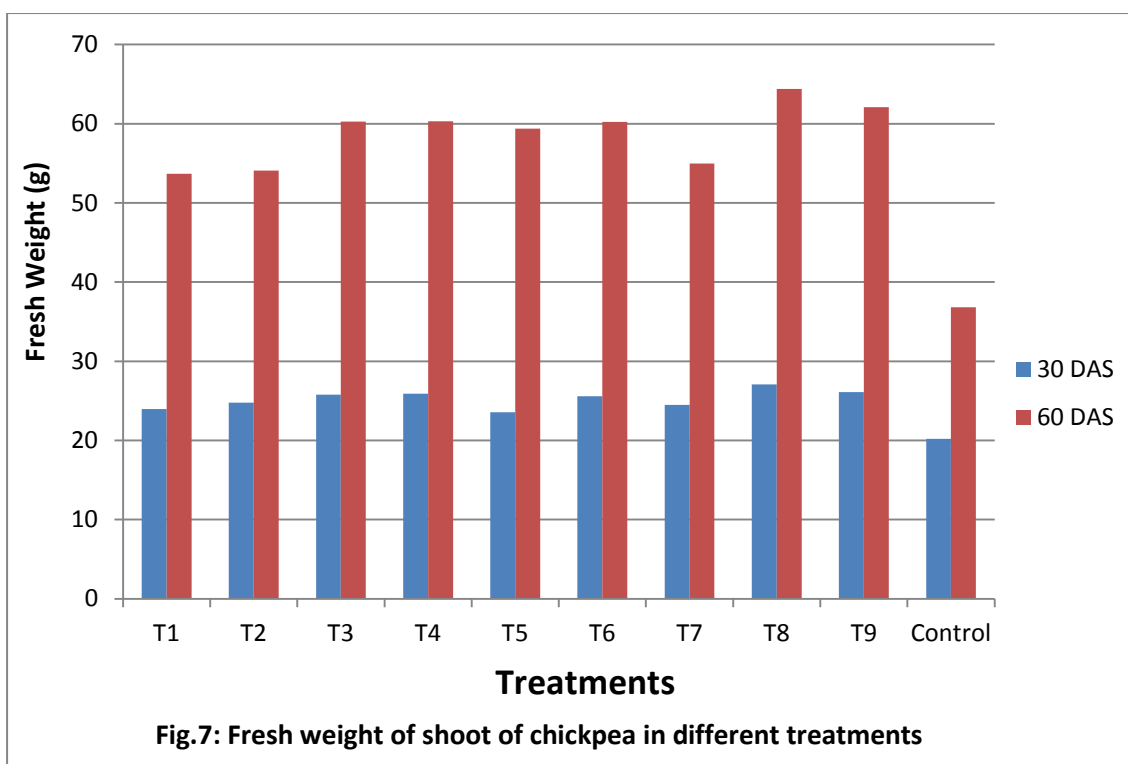


Fig-4: Regression line between disease incidence and relative humidity (%)

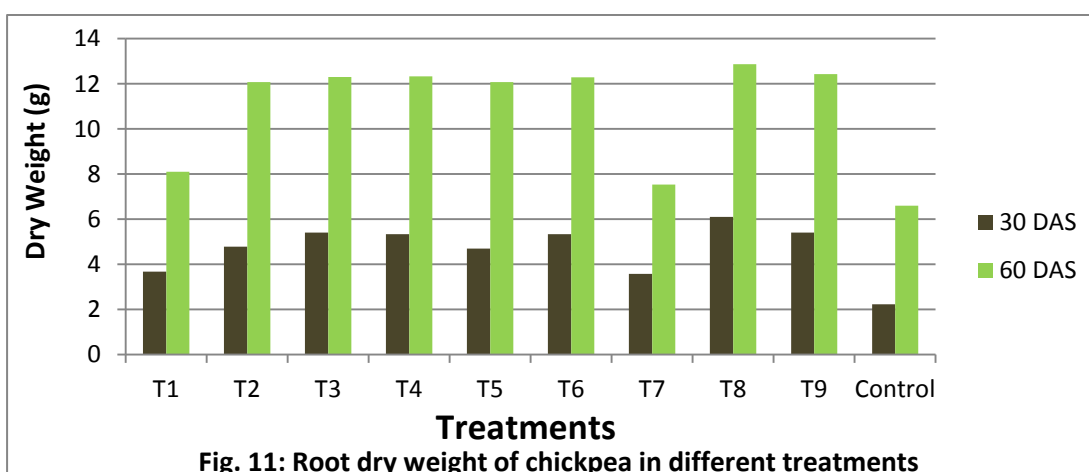
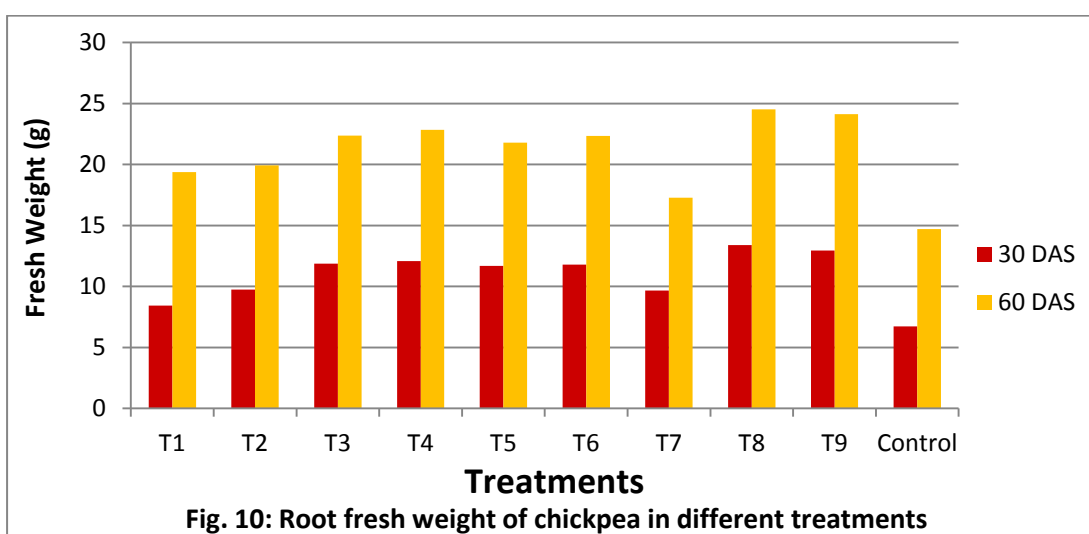
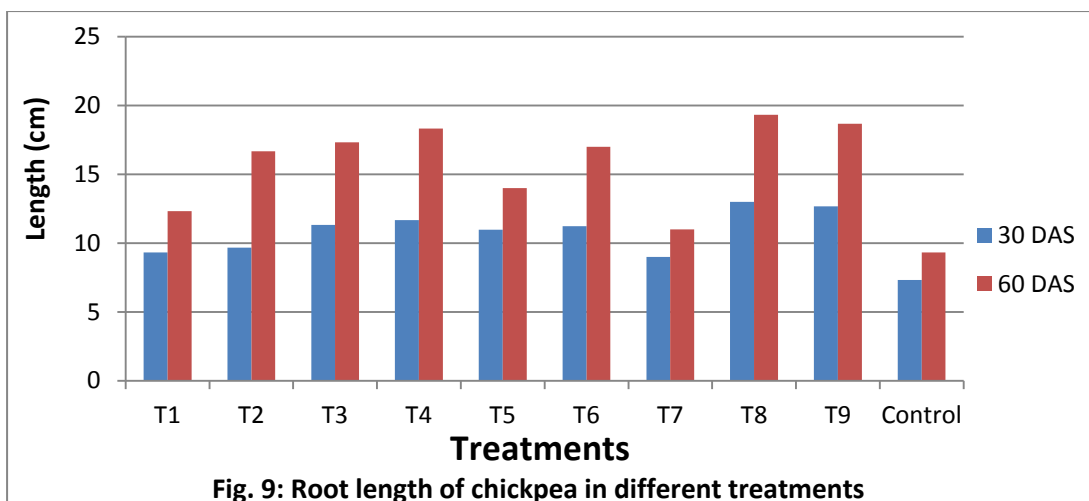




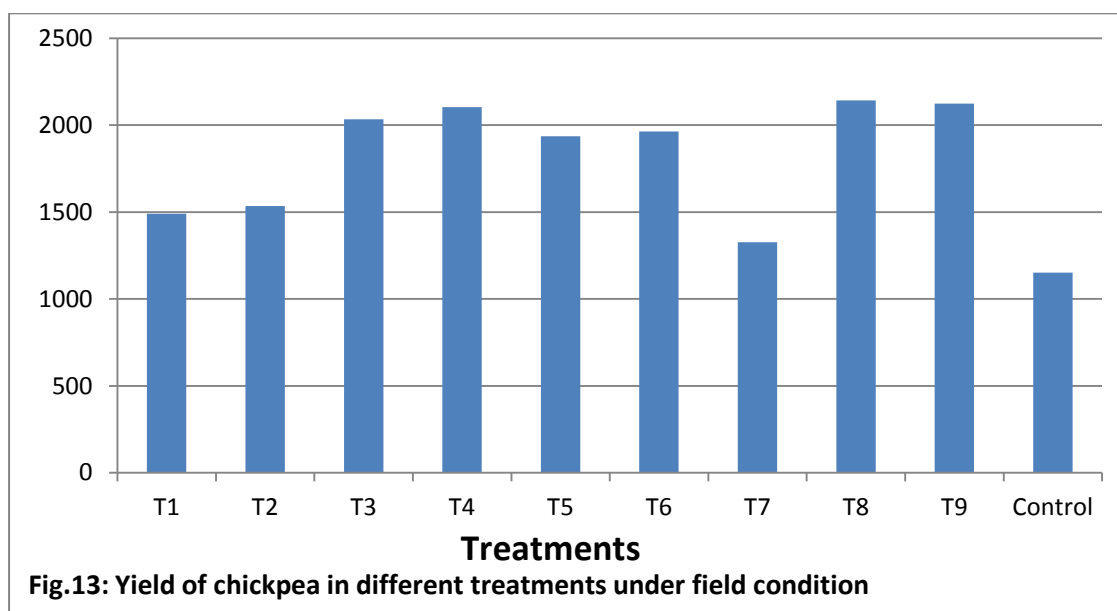
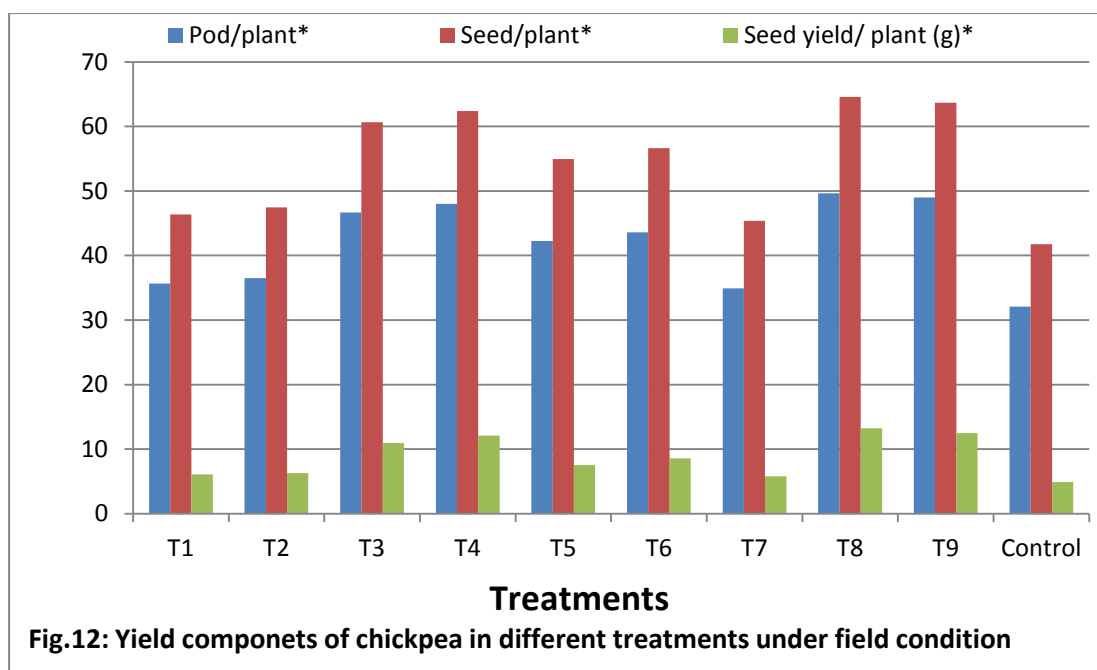
T₁: Seed treatment (ST) by *Bacillus subtilis* @ 2g/kg of seed, **T₂:** ST by *B. subtilis* @ 4g/kg of seed, **T₃:** ST by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, **T₄:** ST by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, **T₅:** Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, **T₆:** Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, **T₇:** Thirum + Carbendazim @ (2:1) (0.3%), **T₈:** *Trichoderma* + Carboxin @ (4:1g)/kg, **T₉:** *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, **T₁₀:** Untreated control.



T₁: Seed treatment (ST) by *Bacillus subtilis* @ 2g/kg of seed, T₂: ST by *B. subtilis* @ 4g/kg of seed, T₃: ST by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, T₄: ST by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, T₆: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, T₇: Thirum + Carbendazim @ (2:1) (0.3%), T₈: *Trichoderma* + Carboxin @ (4:1g)/kg, T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, T₁₀: Untreated control.



T₁: Seed treatment (ST) by *Bacillus subtilis* @ 2g/kg of seed, T₂: ST by *B. subtilis* @ 4g/kg of seed, T₃: ST by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, T₄: ST by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, T₅: Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, T₆: Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, T₇: Thirum + Carbendazim @ (2:1) (0.3%), T₈: *Trichoderma* + Carboxin @ (4:1g)/kg, T₉: *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, T₁₀: Untreated control.



T₁: Seed treatment (ST) by *Bacillus subtilis* @ 2g/kg of seed, **T₂:** ST by *B. subtilis* @ 4g/kg of seed, **T₃:** ST by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @250 g/ha after 30 and 45 days of sowing, **T₄:** ST by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing, **T₅:** Soil + foliar drench of *B. subtilis* @ 250g/ha after 30,45 and 60 days of sowing, **T₆:** Soil + foliar drench of *B. subtilis* @ 500g/ha after 30,45 and 60 days of sowing, **T₇:** Thirum + Carbendazim @ (2:1) (0.3%), **T₈:** *Trichoderma* + Carboxin @ (4:1g)/kg, **T₉:** *Trichoderma* + Ammonium Molybdate @ (4:1g)/kg, **T₁₀:** Untreated control.

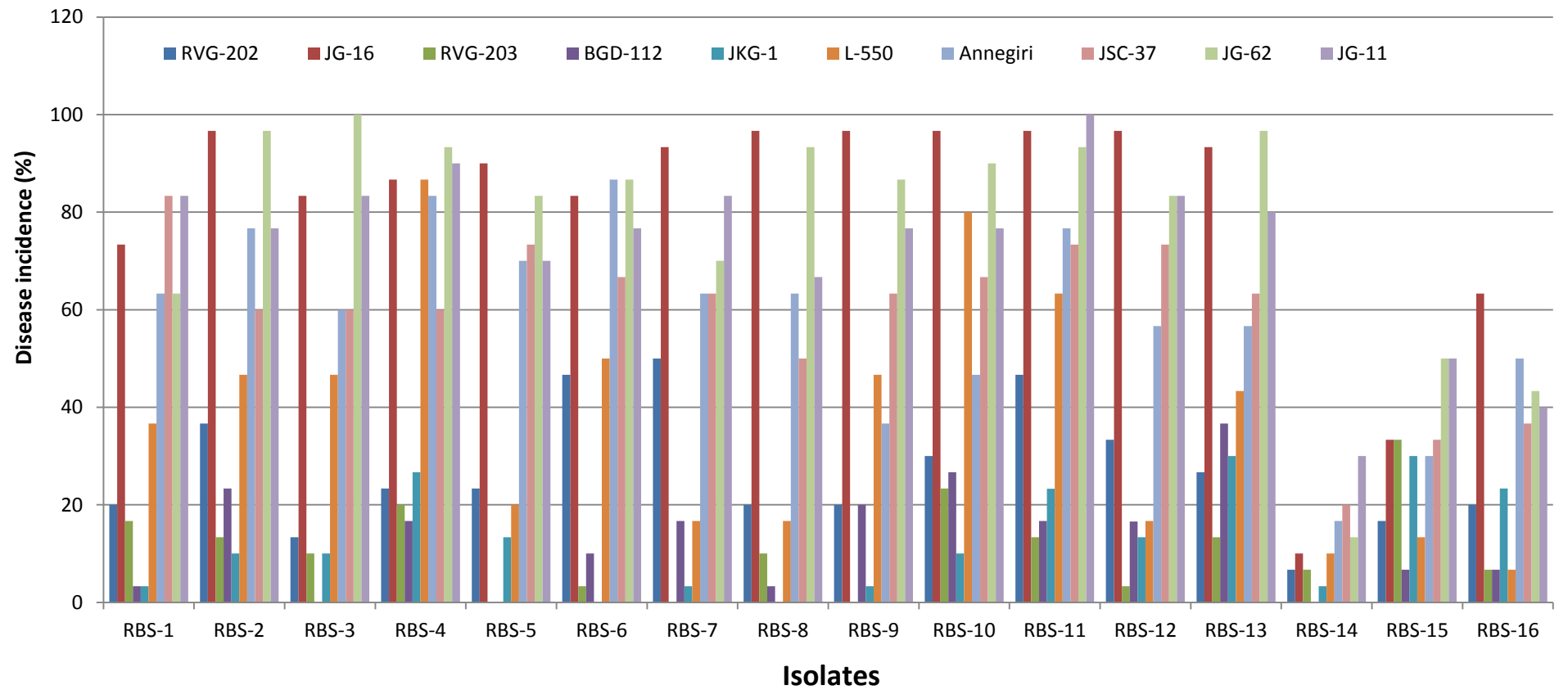


Fig. 1: Disease incidence in mortality percentage of different varieties of chickpea due to *Rhizoctonia bataticola*

CHAPTER-V

DISCUSSION

In Indian agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Chickpea is an important pulse crop of the Indian subcontinent.

Recent reports indicated that dry root rot is emerging as a potential threat to chickpea reduction (Sharma *et al.*, 2016). Several workers have reported the pathogen associated with dry root rot of chickpea.

Survey of dry root rot of chickpea around Sehore district indicated that the disease is prevalent in the region maximum up to 39.30 per cent at some places at flowering and podding stage causing death of the plant. *Rhizoctonia bataticola* is a soil inhabiting organism capable of infecting chickpea at any crop stage, but most commonly infects chickpea at post reproduction stage (Sharma and Pandey, 2013; Khan, 2012). The disease has also been reported from parts of other states of India ranging from 5 to 50 % (Ghosh *et al.*, 2013).

Dry root rot affected plants were identified based on symptoms like drying and withering of plants and when these plants were pulled out showed blackening of tap root and devoid of lateral and finer roots and shredding of bark which comes out in the form of flakes. These symptoms were similar to earlier reports of Singh and Agrawal (2002).

The pathogen isolated from the root region was purified by single hyphal tip method and maintained on PDA for further studies. Based on morphological characters the pathogen was identified as *R. bataticola* Butler. Similar reports were observed by Barnett and Hunter (1972) and Sajana *et al.* (2004).

Pathogenicity of the fungus was established by inoculation and reisolation from artificially infected plants. In the present study, soil inoculation method was followed to establish the disease. Several workers, found soil inoculation method as the most suitable in establishing the disease caused by *R. bataticola* (Kataria and Gaur, 2007).

Based on mycelial growth on PDA sixteen isolates were placed into two groups. The first group included three isolates (RBS-2, RBS-7 and RBS-9) having aerial growth whereas, thirteen isolates (RBS-1, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-10, RBS-11, RBS-12, RBS-13, RBS-14, RBS-15 and RBS-16) comprised of second group with suppressed growth. Aghakhani and Dubey (2009) and Verma and Pathe (2013) also categorized the isolates on the basis of mycelial growth into two groups as fluffy and submerged. Gupta *et al.*, (2012) also reported mycelial growth as submerged, partially submerged and fluffy. Sundravadana *et al.*, (2012) reported variability among the eleven isolates of *R. bataticola* from different pulse crops (redgram, greengram, cowpea, soybean, blackgram). They categorized it into three groups viz., linear, fluffy and linear at the end with fluffy growth at the centre.

Colony colour is divided into six groups. First dark brown group included RBS-1, RBS-4, RBS-5, RBS-6 and RBS-13. Other groups, Whitish grey to black colour of mycelium was observed in four isolates viz., RBS-2, RBS-7, RBS-15 and RBS-16. Isolates RBS-3, RBS-11 and RBS-14 were grey to black coloured whereas, isolate RBS-8 showed brown to black. Only one isolate RBS-9 showed whitish grey colour. Isolates RBS-10 and RBS-12 showed black coloured colonies. Aghakhani and Dubey (2009) also reported variation in colony colour from white to dull white, creamy, grey and black. Sharma *et al.* (2012) found variation in colony colour like light black to black and light grey to grey. Present study confirms the observations made by Gupta *et al.* (2012) who reported colony colour varied from dark black to grayish. Verma and Pathe (2013) observed colony colour as dark brown.

On the basis of average growth per day isolates were placed into three groups viz., slow growth rate, medium growth rate and fast growth rate. Isolates RBS-4, RBS-8, RBS-13 and RBS-15 exhibited slow growth and isolates RBS-1, RBS-3, RBS-5, RBS-6, RBS-7, RBS-10, RBS-12, RBS-14 and RBS-16 showed medium growth. Rest of the isolates viz., RBS-2, RBS-9 and RBS-11 attained the fast growth. Aghakhani and Dubey (2009) also reported variable radial growth twenty three isolates of *R. bataticola* causing dry root rot of chickpea.

Microscopic observation of *R. bataticola* isolates showed variation in size. RBS-16 (72.84 x 47.06 µm) isolate produced largest sclerotia, while the smallest sized sclerotia were recorded in RBS-7 (36.16 x 21.96 µm) isolate. Four isolates were produced large size sclerotia, two isolates produced medium size sclerotia and ten isolates produced small size sclerotia. Isolates RBS-1, RBS-7, RBS-14 and RBS-15 produced dark brown coloured sclerotia whereas, isolates RBS-2, RBS-3, RBS-4, RBS-5, RBS-6, RBS-8, RBS-9, RBS-10, RBS-11, RBS-12, RBS-13 and RBS-16 produced black coloured sclerotia. Verma and Pathe (2013) found the size of sclerotia varied from 72.2 – 117.2 × 57.1 -106.5 µm. Different workers had given the wide range of dimensions of sclerotia size in different crops. Dhingra and Sinclair (1973) found sclerotia from root isolates ranged from 122-188 × 77-127 µm.

On the basis of number of sclerotia per microscopic field sixteen isolates of *R. bataticola* were divided into three groups. First group with (5 to 20 sclerotia) minimum, medium (21-40 sclerotia) and third maximum (40 < sclerotia). Verma and Pathe (2013) grouped *R. bataticola* isolates according to number of sclerotia/ micro field into three categories of sparse, medium and abundant. The number of sclerotia ranged from 9.70 to 22.2.

Ten chickpea cultivars viz., RVG-202, JG-16, RVG-205, BGD-112, JKG-1, L-550, Annagiri, JSC-37, JG-62 and JG-11 selected for assessing pathological variability resulted into varied response against *R. bataticola* isolates. Three varieties namely RVG-205, BGD-112 and JKG-1 were found moderately resistant. Two varieties namely RVG-202 and L-550 were found susceptible and rest five varieties namely JG-16, Annagiri, JSC-37, JG-62 and JG-11 were found highly susceptible against the disease. Cultivars RVG-202, JG-16 and L-550 were resistant to RBS14, RVG-205 was resistant to RBS3, RBS5, RBS6, RBS7, RBS8, RBS9, RBS12, RBS14 and RBS16. BGD-112 was resistant to RBS1, RBS3, RBS5, RBS6, RBS8, RBS14, RBS15 and RBS16. JKG-1 was resistant to RBS1, RBS2, RBS3, RBS6, RBS7, RBS8, RBS9, RBS10 and RBS14. Pathogenic variation in *R. bataticola* have also been reported from different parts of country (Lekhraj *et al.*, 2012, Sharma *et al.*, 2012), which supports the present findings from Madhya Pradesh.

The development of dry root rot intensity was studied in relation to various climatic parameters. Maximum temperature and minimum temperature, were found to have a significant positive correlation with dry root rot of chickpea ($r=0.80$ and 0.66 respectively) and relative humidity was found to have a significant negative correlation with dry root rot of chickpea ($r=-0.38$). Under controlled environment. Sharma and Pande (2013) observed that *R. bataticola* infected chickpea plants had caused dry root rot faster at 35°C with soil moisture levels less than or equal to 60%. Further, *R. bataticola* propagules were negatively correlated with high soil moisture and positively with less soil moisture content.

High day temperature above 30°C and dry soil conditions at flowering and podding stages rapidly increase the severity of the dry root rot disease on chickpea (Gurha *et al.*, 2003).

The use of antagonistic bacteria is reported as a powerful strategy to suppress soil-borne pathogens due to their ability to colonize the rhizosphere and ability to antagonize the pathogen by multiple modes of action. *Bacillus* spp. is recognized as safe biocontrol agents specifically as seed protectants and antifungal agent (Asaka and Shoda, 1996, Stein, 2005). Present investigation shows that significantly less mortality due to *R. bataticola* was observed in *Trichoderma* seed treatment along with carboxin and ammonium Molybdate (14.58 and 16.37%, respectively).

The reduction of *Fusarium* infection in field following the application of *Bacillus* spp. has been reported by many workers (Khan *et al.*, 2001; Crane *et al.*, 2013). Chickpea seedling were significantly protected against *Fusarium* wilt by *Bacillus* spp. Rb29 and reduced the disease severity and incidence caused by *F. oxysporum* f. sp. *ciceri* (Zaim *et al.*, 2013, Smitha *et al.*, 2017). The populations of secondary metabolites by *Bacillus* spp. including antibiotics of low molecular weight and natural cyclic peptides have been related to the inhibition of many fungal species (Romero *et al.*, 2007). The *B. subtilis* has already been reported to produce five different lipopeptide antibiotics viz., surfactin, iturin A, iturin D, bacillomycin D and fengycin D (Smitha *et al.*, 2015).

Kumar *et al.* (2013) also found that *T. harzianum*, *T. Viride* and *T. atroviride*, and one bacterium, *Bacillus subtilis* reduced root rot incidence in a susceptible moth bean up to 69.0, 65.0, 62.5 and 58.3%, respectively, as compared to control where 100 % mortality was observed.

Moreover, seed treatment (4g/kg seed) along with foliar drench at 30, 45 and 60 days with *Bacillus* was more effective than seed treatment alone. It shows that seed treatment with the antagonists might not be efficient enough to protect the root infection courts. Loper *et al.* (1984) demonstrated that colonization of roots by selected microbes placed on seeds can occur in a log-normal distribution; consequently, microbial populations decrease on the root as the root moves away from the court of microbial activity. This relates to results by Herva's *et al.* (1997), who demonstrated that colonization of chickpea roots and protection against Fusarium wilt were more efficient in a transplant system as opposed to a seed treatment. Similarly, Landa *et al.* (1997a) demonstrated that seed and soil treatment was more efficient for establishing species of *Bacillus* for biological control than either seed or soil treatment alone. Recently Smitha *et al.* (2017) found that talc based formulation of *B. subtilis* CaB5@10g/kg seed + basal application of CaB5 @ 2.5kg/ha followed by soil application of CaB5 at 45 DAS reduced the wilt and root rot incidence in chickpea by 72.0 and 61.1 per cent respectively.

In recent years, there has been a paradigm shift with evolving interest in the use of biological approaches, as an alternative to chemical fertilizers, to improve crop productivity and plant resistance to phytopathogens (Rosas *et al.* 2009; Dasatger *et al.*, 2011). Among bacteria, species belonging to *Bacillus* play an important role in the agro ecosystems as this bacterium helps in improving plant growth and nutrient assimilation (Ramirez and Kloepper, 2010).

In the present study all the bio-agent treatments significantly increased the plant vigour, shoot fresh and dry weight as compare to control. Among the treatment seed treatment with highest plant vigour was noticed in *Trichoderma* + ammonium molybdate treatment followed by *Bacillus* treatments. Similarly highest shoot fresh and dry weight was noticed in

Trichoderma +carboxin seed treatment followed *Trichoderma* + molybdate and seed and soil application of *Bacillus*. Das (2006) observed that seeds treated with formulation of *T. harzianum* + tac + ostmatic formulation enhanced the plant vigour. Madhy *et al.* also reported enhancement in plant growth parameters *i.e.* shoot and root fresh and dry weights in biological agents, *B. thuringiensis* and *T. harzianum* treated plants as compare to control. Growth improvement of *Trichoderma* treated plant could be due to synergic activity of *Trichoderma* strain on the host plant. *Trichoderma* might colonize around the root and increased root biomass and helped to increase the availability of nutrients (Vinale *et al.*, 2008).

Stefan *et al* (2010) observed that *Bacillus pumilus* Rs3 promoted plant growth. Karimi *et al* (2012) reported significant increase in length of shoot and root of chickpea plants by soil and seed treatments of *B. subtilis* B28 and *P. aueruginosa* P12 isolates. Plant growth-promotion by rhizosphere *Bacilli* is a consequence of combined effects of different mechanisms such as production of 3-hydroxy-2-butanone (acetoin), 2,3- butanedio, IAA, phytase, lipopeptide, poluketide, siderophore and induced systemic resistance (Koumoutsis *et al.*, 2004; Idriss *et al.*, 2007; Zhao *et al.*, 2011).

All treatments significantly increased the root length, fresh and dry weight of root. However, significantly higher was noticed in *Trichoderma* + corboxin and *Trichoderma* + molybdate seed treatment followed by and seed and soil applications of *Bacillus*. Suresh *et al.* (2011) also reported induction of shoot and root length of chickpea up to 10-15% by *T. viridi* and *T. harzianum*. Improved dry weight of roots by three strains of *Trichoderma* (T.h4, T.v7 and T.a9) were reported by Entesari, (2013). However, in 2000 Harman established that *Trichoderma* spp. are opportunistic plant colonizers that effect plant growth by promoting abundant and healthy roots, possibly due to production of growth hormones.

In the present study seed, soil and drenching of *B. subtilis* increased the root length and weight of chickpea plants as compare to untreated control. This is in consonance with findings of Zhao *et al.* (2011), wherein they reported that inoculation of *B. subtilis* GY-IVI increased root biomass, total

root length and total root volume of plants involving role of IAA like compounds.

Significantly higher numbers of nodules, fresh and dry weight of nodules were observed all the treatments except T1 (*Bacillus subtilis* @ 2g/kg of seed) as compared to control and treatments were at par to each other. Patel *et al.* (2012) reported increase in plant height, fresh and dry weight, no. of nodules of chickpea by *Trichoderma* seed treatment. Similarly, Increase in nodulation by *Bacillus pumilus* Rs3 also reported by Stefen *et al.* (2010).

Some investigators reported that the increased growth response caused by *Trichoderma* isolates resulted in large increase in root area and root lengths and may be related to the effect on root system. These results were similar to the results revealed by (Yedidia *et al.*, 2001) who showed that treatment of cucumber plants in soil with *T. harzianum* (T-203) resulted in large increase in root area and cumulative root lengths and significant increase in dry weight, shoot length and leaf area over that of the untreated control. In addition, Harman (2000) established that *Trichoderma* spp. are opportunistic plant colonizers that affect plant growth by promoting abundant and healthy plant roots, possibly via the production or control of plant hormones. Increased growth response has been demonstrated by several other investigators (Altomare *et al.*, 1999; Anusuya and Jayarajan, 1998).

Significantly maximum average no. of pod/plant, seed/plant and yield/ha obtained in T8 (*Trichoderma*+Carboxin @ (4:1g)/kg) which was at par with T9 (*Trichoderma* + Amm. Molybdate @ (4:1g)/kg) and T4 (Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing). Increase in yield by *Trichoderma* seed treatment results are in agreement with Pandey *et al.* (2016) who observed disease control and yield enhancement with *T. viride* followed by *T. harzianum*.

According to Dillfiza *et al* (2017) the strain *B. subtilis* NUU4 proved significant plant growth promotion capabilities, improved symbiotic performance of host plant with rhizobia, and promoted yield under saline soil as compared to untreated control plants under field condition.

CHAPTER-VI

SUMMARY, CONCLUSION AND SUGGESTIONS FOR FUTURE WORK

❖ Summary

The present investigation on variability and management of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea was carried out during *rabi* 2013-14, at Department of Plant Pathology and field of R. A. K. College of Agriculture, Sehore (M.P.), R.V.S.K.V.V. Gwalior (M.P.)

- ✓ Survey of chickpea fields for incidence of dry root rot disease at flowering and pod formation stage was conducted in five blocks of Sehore i.e. Sehore, Shyampur, Ichhawar, Ashta and Nasrulaganj where severe incidence of diseases was reported earlier. The disease incidence was 29.28%, 28.45%, 29.25%, 33.16% and 23.58% in Sehore, Shyampur, Ichhawar, Ashta and Nasrulaganj, respectively.
- ✓ The dry root rot symptoms observed were mainly brittleness of tap root and presence of minute sclerotia. These symptoms separated this disease from the so called wilt complex as termed by various workers in India.
- ✓ Isolates of *R. bataticola* exhibited variability in mycelium growth. First group included these isolates (RBS--1, RBS--3, RBS--4, RBS--5, RBS--6, RBS--8, RBS--10, RBS--11, RBS--12, RBS--13, RBS--14, RBS--15 and RBS—16) having aerial growth and second group having aerial growth. (RBS--2, RBS--7 and RBS—9).
- ✓ On the basis of mycelium colour *R. bataticola* isolates were grouped into six viz., dark brown group (RBS-1, RBS-7, RBS-14 and RBS-15), whitish grey to black colour (RBS-2, RBS-7, RBS-15 and RBS-16), grey to black colour (RBS-3, RBS-11 and RBS-14), brown to black (RBS-8), whitish grey (RBS-9) and black (RBS-10 and RBS-12).
- ✓ Among the sixteen isolates of *R. bataticola*, isolates RBS-2, RBS-9 and RBS-11 showed fast mycelium growth.
- ✓ RBS-16 isolate produced largest sclerotia, while the smallest sized sclerotia was recorded in RBS-7 isolate.

- ✓ The number of sclerotia/microscopic field 10 X was maximum in RBC-12 isolate, while minimum number was observed in RBC-10.
- ✓ Dark brown coloured sclerotia produced in four isolates (RBS-1, RBS-7, RBS-14 and RBS-15) and while black colour sclerotia in twelve.
- ✓ The development of dry root rot intensity was studied in relation to various climatic parameters. Maximum temperature and minimum temperature were found to have a significant positive correlation with charcoal rot of chickpea ($r=0.80$ and 0.66 respectively) and relative humidity was found to have a significant negative correlation with dry root rot of chickpea ($r=-0.38$).
- ✓ Significantly highest germination per cent and lowest incidence of dry root rot were recorded in (T8) *Trichoderma*+Carboxin followed by (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing.
- ✓ Evaluation of *Bacillus* strains for growth promoting ability (GPA) of chickpea. Among the treated treatment, (T8) *Trichoderma*+Carboxin was obtained highest plant vigour followed by (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing as compared to other treatment and control.
- ✓ The highest fresh and dry shoot weight was observed in (T8) *Trichoderma*+Carboxin at 30 and 60 days after sowing followed by (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing.
- ✓ The highest root length after 30 and 60 days recorded in (T8) *Trichoderma*+Carboxin which was at par with (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar

application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing.

- ✓ Treatment (T8) *Trichoderma*+Carboxin significantly increase the fresh and dry root weight at 30 and 60 days after sowing which was at par with (T9) *Trichoderma*+Amm. Molybdate.
- ✓ Highest number of nodules, nodule fresh and dry weight were observed in treatment T8 (*Trichoderma*+Carboxin), which was at par with treatments T9 (*Trichoderma*+Amm. Molybdate) and T4 (Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing) after 30 days of sowing.
- ✓ Treatments with *Bacillus subtilis*, *Trichoderma* strain and chemical combinations and chemicals increased yield and yield attributing characters. Significantly maximum average numbers of pods were recorded in (T8) *Trichoderma*+Carboxin which was at par with (T9) *Trichoderma* + Amm. Molybdate and (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing.

❖ Conclusion

- ✓ Disease incidence of dry root rot of chickpea is more during flowering and pod formation stage.
- ✓ Studies on sixteen isolates of *R. bataticola* obtained from chickpea around the Sehore districts of Madhya Pradesh showed considerable variability in morphological characters and may be grouped into 2-3 pathotypes.
- ✓ Present studies conducted on various aspects of variability in the isolates of *R. bataticola* led to be conclusion that the isolates having dark brown mycelium (5 isolates), medium growth rate (9 isolates), suppressed mycelial growth (13 isolates), black coloured sclerotia (12 isolates), minimum no. of sclerotia (8 isolates) and sclerotia small size (7 isolates).
- ✓ Maximum temperature, minimum temperature, rainfall (mm) and no. of rainy days were found to have a significant positive correlation with

charcoal rot of chickpea and relative humidity was found to have a significant negative correlation with dry root rot of chickpea.

- ✓ As an evaluation of different treatments for growth promoting ability (GPA) of chickpea, it could be concluded that treatment (T8) *Trichoderma*+Carboxin was highest increased growth of chickpea plant followed by (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing was best.
- ✓ The highest yield/ha was obtained in (T8) *Trichoderma*+Carboxin which was at par with (T9) *Trichoderma*+Amm. Molybdate, (T4) Seed treatment by *B. subtilis* @ 4g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing and (T3) Seed treatment by *B. subtilis* @ 2g/kg and foliar application of *B. subtilis* @ 250 g/ha after 30 and 45 days of sowing as compared to rest of treatments and control.

❖ **Suggestion for future work**

- Variations in different isolates of *R. bataticola* on their molecular basis need to be worked out.
- Morphological, physiological and biochemical factors responsible for disease resistance needs investigation. The mechanism of resistance is different identified cultivars should be find out.
- There is need to verify the resistance of genotypes under multi locations to know the wide base of resistance.

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APPENDICES

Analysis of variance for Table-6

SV	DF	SS	MSS	Fcal	Ftab
Isolates	15	46061.06	3070.737	15.74862	2.07
Varieties	9	199036.9	22115.21	113.4204	2.71
Inter	135	51240.87	379.562	1.946627	1.00
Error	320	62395.03	194.9845		
Total	479	358733.9			

Analysis of variance for Table-8

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	1.032735	0.516367	2.675714	3.55
Treatments	9	2474	274.8889	1424.42	2.34
Error	19	3.666677	0.192983		
Total	29	2478.699			

Analysis of variance for Table-9

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	377866.1	188933	3.441083	3.55
Treatments	9	24855498	2761722	50.29991	2.34
Error	18	988291.9	54905.11		
Total	29	26221656			

Analysis of variance for Table-10 (Fresh weight of shoot at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.294	0.147	0.370277	3.55
Treatments	9	98.912	10.99022	27.68318	2.34
Error	18	7.146	0.397		
Total	29	106.352			

Analysis of variance for Table-10 (Fresh weight of shoot at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	5.408667	2.704333	4.221368	3.55
Treatments	9	1645.882	182.8757	285.4625	2.34
Error	18	11.53133	0.64063		
Total	29	1662.822			

Analysis of variance for Table-10 (Dry weight of shoot at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.172667	0.086333	0.439822	3.55
Treatments	8	166.7803	20.84754	106.207	2.34
Error	16	3.140667	0.196292		
Total	26	170.0937			

Analysis of variance for Table-10 (Dry weight of shoot at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	2.162667	1.081333	0.864318	3.55
Treatments	8	279.5987	34.94983	27.93566	2.34
Error	16	20.01733	1.251083		
Total	26	301.7787			

Analysis of variance for Table-11 (Root length at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.2	0.1	0.050279	3.55
Treatments	9	100.8	11.2	5.631285	2.34
Error	18	35.8	1.988889		
Total	29	136.8			

Analysis of variance for Table-11 (Root length at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	3.8	1.9	0.927667	3.55
Treatments	9	330.5333	36.72593	17.93128	2.34
Error	18	36.86667	2.048148		
Total	29	371.2			

Analysis of variance for Table-11 (Fresh weight of root at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.480667	0.240333	1.144646	3.55
Treatments	9	121.4697	13.49663	64.281	2.34
Error	18	3.779333	0.209963		
Total	29	125.7297			

Analysis of variance for Table-11 (Fresh weight of root at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	1.152	0.576	0.916873	3.55
Treatments	9	264.388	29.37644	46.76123	2.34
Error	18	11.308	0.628222		
Total	29	276.848			

Analysis of variance for Table-11 (Dry weight of root at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.164667	0.082333	0.66915	3.55
Treatments	8	93.77633	11.72204	95.26888	2.34
Error	16	1.968667	0.123042		
Total	26	95.90967			

Analysis of variance for Table-11 (Dry weight of root at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.104667	0.052333	0.570132	3.55
Treatments	8	463.4613	57.93267	631.1321	2.34
Error	16	1.468667	0.091792		
Total	26	465.0347			

Analysis of variance for Table-12 (No. of nodules at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	8.266667	4.133333	1.516304	3.55
Treatments	9	1208.033	134.2259	49.24049	2.34
Error	18	49.06667	2.725926		
Total	29	1265.367			

Analysis of variance for Table-12 (No. of nodules at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	20.6	10.3	0.637698	3.55
Treatments	9	579163.5	64351.5	3984.156	2.34
Error	18	290.7333	16.15185		
Total	29	579474.8			

Analysis of variance for Table-12 (Fresh weight of nodule at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	62.06667	31.03333	2.44357	3.55
Treatments	9	3488.3	387.5889	30.51881	2.34
Error	18	228.6	12.7		
Total	29	3778.967			

Analysis of variance for Table-12 (Fresh weight of nodule at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	180.6	90.3	2.339603	3.55
Treatments	9	14818.97	1646.552	42.66088	2.34
Error	18	694.7333	38.5963		
Total	29	15694.3			

Analysis of variance for Table-12 (Dry weight of nodule at 30 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.654	0.327	2.821668	3.55
Treatments	9	379.288	42.14311	363.651	2.34
Error	18	2.086	0.115889		
Total	29	382.028			

Analysis of variance for Table-12 (Dry weight of nodule at 60 DAS)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	0.104	0.052	0.020119	3.55
Treatments	9	4174.245	463.805	179.45	2.34
Error	18	46.52267	2.584593		
Total	29	4220.872			

Analysis of variance for Table-13 (Pod/plant)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	24.640607	12.3203033	2.4964912	3.55
Treatments	9	1158.8601	128.76223	26.091385	2.34
Error	18	88.83086	4.93504778		
Total	29	1272.3315			

Analysis of variance for Table-13 (Seed/plant)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	7.77	3.89	1.09	3.55
Treatments	9	1403.38	155.93	43.90	2.34
Error	18	63.93	3.55		
Total	29	1475.09			

Analysis of variance for Table-13 (Seed yield/ plant)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	3.81	1.90	1.37	3.55
Treatments	9	307.02	34.11	24.48	2.34
Error	18	25.08	1.39		
Total	29	335.91			

Analysis of variance for Table-13 (Yield/ha)

SV	DF	SS	MSS	Fcal	Ftab
Replications	2	6142.82	3071.41	0.37	3.55
Treatments	9	3668664.21	407629.36	48.69	2.34
Error	18	150706.18	8372.57		
Total	29	3825513.21			

VITA

The author of this thesis Jagdish Prasad S/o Shri Kuwarji was born on 03rd July, 1983 at Vill.Haidarganj Post-Muskara Tehsil + District - Sehore (M.P.).

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